

The Closest Reachable Vector

From Exact Solutions to Best Approximations

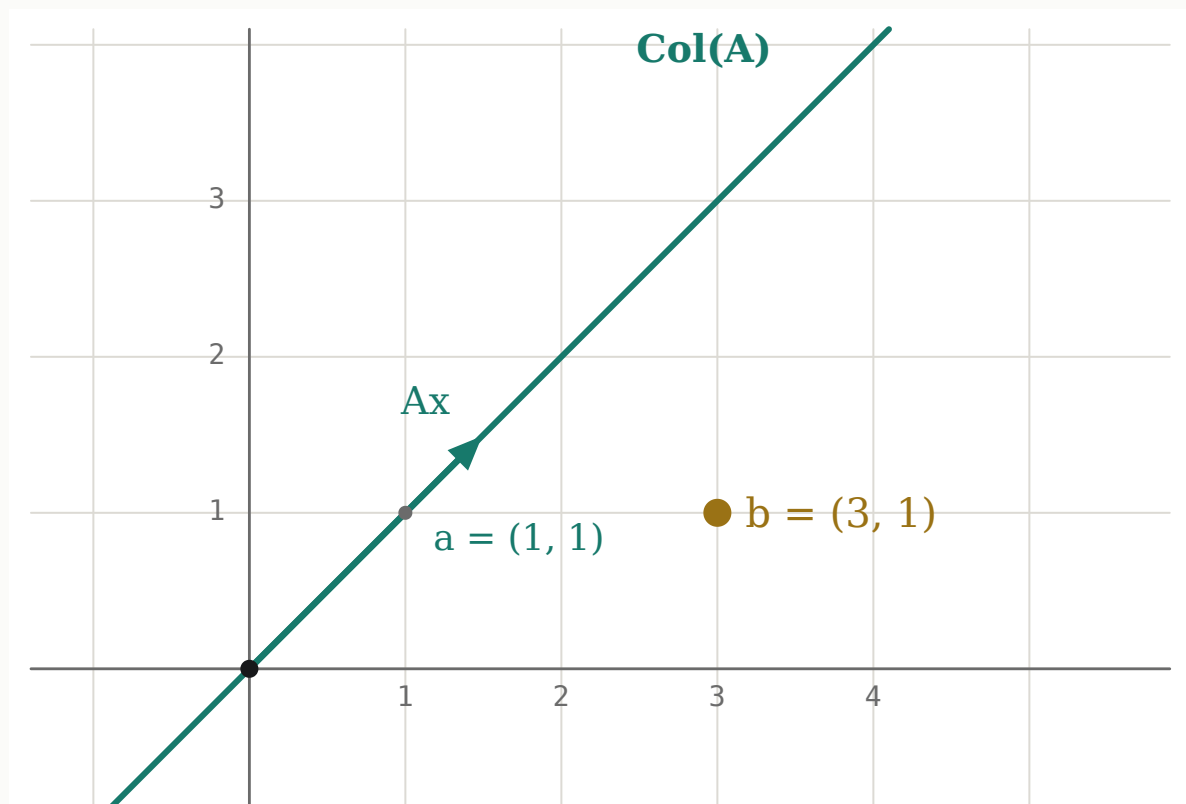
$$Ax = x_1 a_1 + \cdots + x_n a_n$$

$A \in \mathbb{R}^{m \times n}$ choose $x \in \mathbb{R}^n$ — parameter space

$a_j, Ax, b \in \mathbb{R}^m$ compare here — output space

Exact solution iff $b \in \text{Col}(A)$.

What if the target is outside the reachable space?



$$a = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad b = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

The Closest Reachable Vector

Which Reachable Vector Would You Choose?

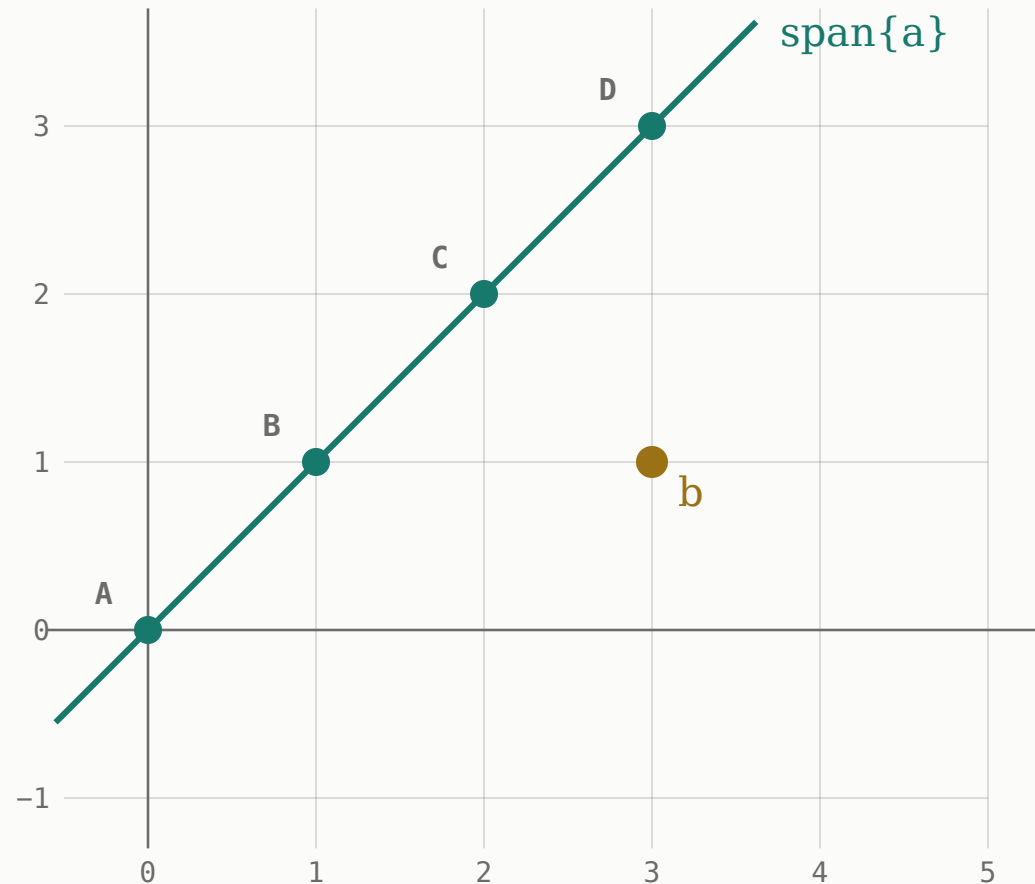
Which point on the line is closest to b ?

(A) (0, 0)

(B) (1, 1)

(C) (2, 2)

(D) (3, 3)



DEFINITION

The Closest Reachable Vector

Difference, Norm, Squared Norm

So far: *representable? unique?— closest* needs a distance.

$$r = u - v$$

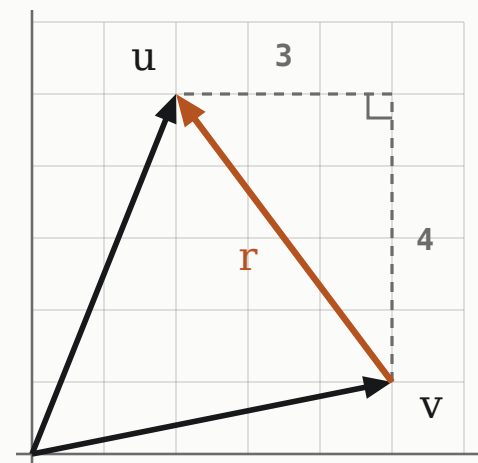
$$\|r\|_2^2 = r^T r = \sum_{i=1}^m r_i^2$$

$$d(u, v) = \|u - v\|_2 = \sqrt{r^T r}$$

$u, v, r \in \mathbb{R}^m$ r = difference vector

$\|r\|_2$ a number ≥ 0 ; $= 0$ iff $r = 0$

$\|r\|_2^2$ a number too, no square root



$$r = (-3, 4)$$

$$\|r\|^2 = (-3)^2 + 4^2 = 25$$

$$\|r\| = \sqrt{25} = 5$$

The Closest Reachable Vector

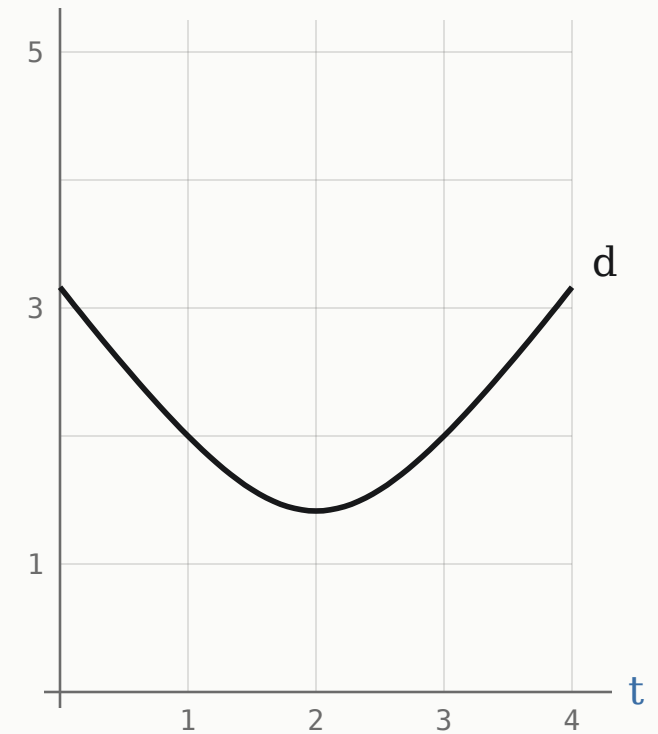
Does Squaring Move the Best Choice?

Candidates $y = (t, t)$ on the same line, $d(t) = \|\mathbf{b} - y\|_2$.

We minimise d^2 instead of d . Does the best t move?

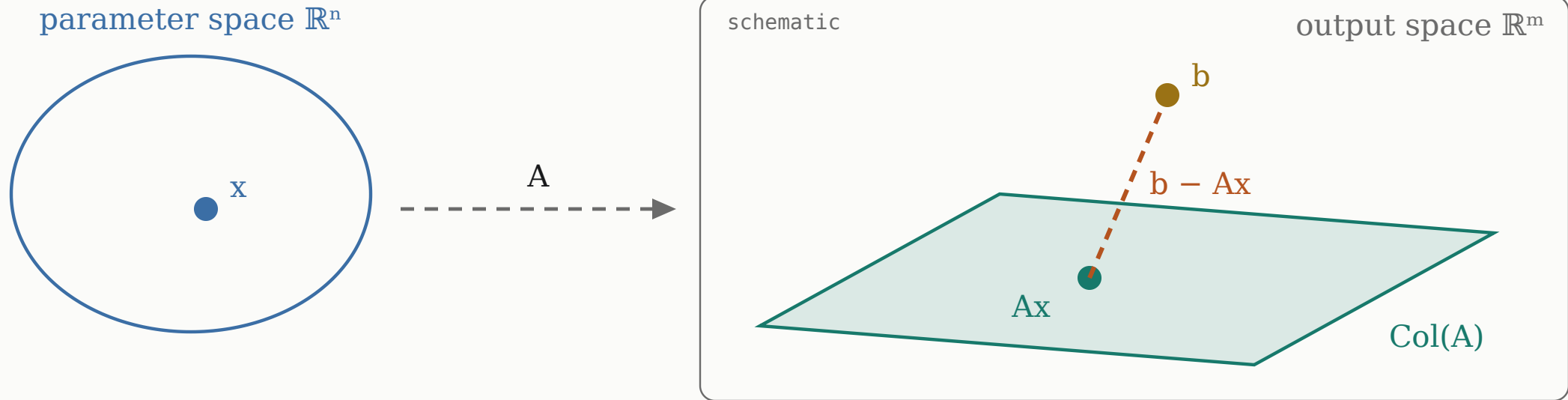
☐ A yes

☐ B no



The Closest Reachable Vector

Today's Roadmap



Find the best output first, then find parameters that produce it; the error is always measured in output space.

1
Judge orthogonality

2
Compute a
projection

3
Set up least squares

4
Gram-Schmidt and
QR

5
A robot step

DEFINITION

Orthogonality and Orthonormal Bases

The Euclidean Inner Product

$$u^T v = \sum_{i=1}^m u_i v_i$$

$u, v \in \mathbb{R}^m$ real; $u^T v$ is a number, not a vector

Lecture 2 wrote $\langle u, v \rangle$

$$u^T v = v^T u$$

$$(c_1 u + c_2 u')^T v = c_1 u^T v + c_2 u'^T v$$

symmetric each $u_i v_i = v_i u_i$

linear $u' \in \mathbb{R}^m$ any other vector, c_1, c_2 any real scalars; slot 2 by symmetry

u	v	$u_i v_i$
$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$	$\times \begin{bmatrix} 3 \\ -1 \end{bmatrix}$	$= \begin{array}{r} 3 \\ -2 \end{array}$
		$\hline u^T v = 1$

Orthogonality and Orthonormal Bases

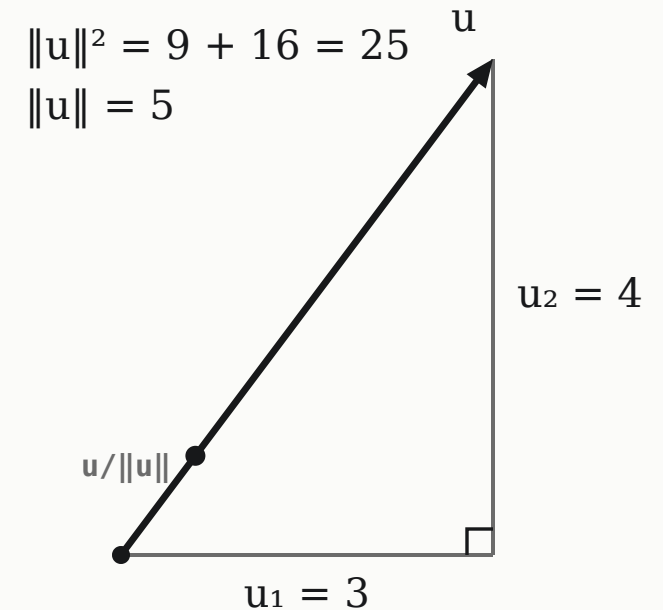
From Inner Product to Length

$$\|u\|^2 = u^T u = \sum_{i=1}^m u_i^2 \geq 0$$

= 0 iff $u = 0$ each term is a square, so 0 forces every $u_i = 0$
same norm the distance $\|r\|_2$, with r for u

$$\|u\| = \sqrt{u^T u}, \quad \left\| \frac{u}{\|u\|} \right\| = 1$$

$\|u\|$ the Euclidean length, a number
unit vector $\|u\| = 1$; dividing by $\|u\|$ needs $u \neq 0$



Orthogonality and Orthonormal Bases

The Dot Product Measures Angle

$$u^T v = \|u\| \|v\| \cos \varphi$$

$\varphi \in [0, \pi]$ the angle between nonzero u, v ; no angle if one is 0, and then $u^T v = 0$

why a rotation R keeps $u^T v$; turn both by R with $Ru = \|u\|e_1$: $u^T v = \|u\|(Rv)_1$

sign $\|u\|\|v\| > 0$, so only $\cos \varphi$ can turn it

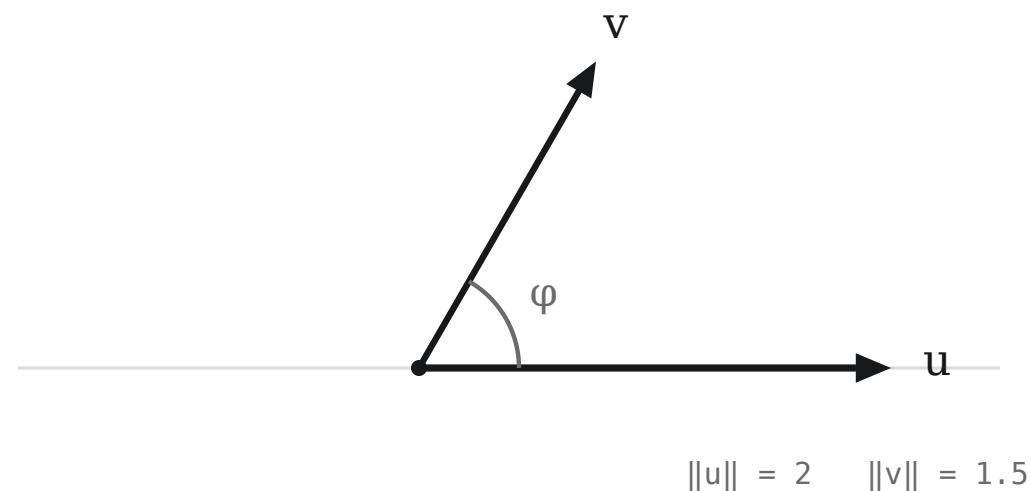
$$\begin{array}{l} \varphi < 90^\circ \\ u^T v > 0 \end{array}$$

$$\begin{array}{l} \varphi = 90^\circ \\ u^T v = 0 \end{array}$$

$$\begin{array}{l} \varphi > 90^\circ \\ u^T v < 0 \end{array}$$

$$\varphi = 60^\circ$$

acute: $u^T v > 0$



Orthogonality and Orthonormal Bases

The Signed Shadow

$$s = \frac{u^T v}{\|u\|} = \|v\| \cos \varphi$$

s a number ($u \neq 0$): the signed component of v along the unit direction of u

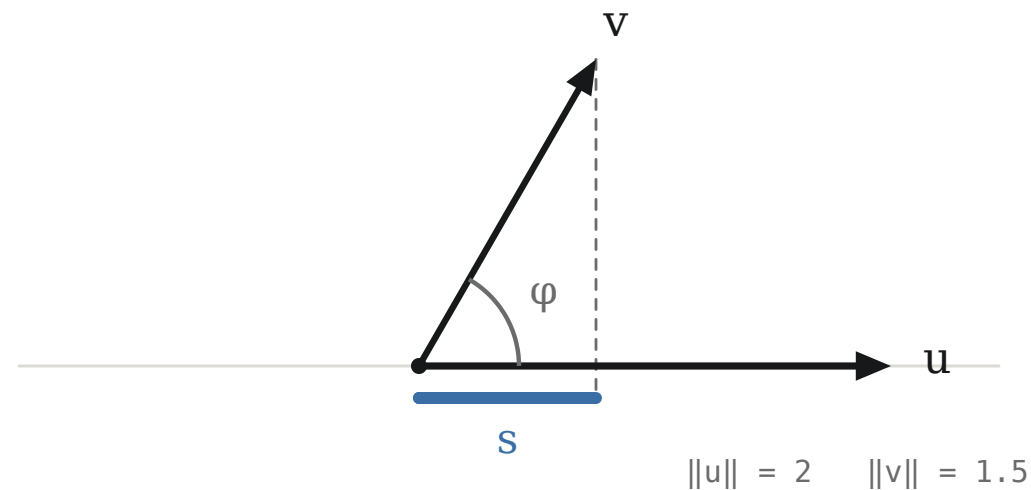
$\|v\| \cos \varphi$ also needs $v \neq 0$: no angle when $v = 0$, though $s = 0$

divisor $\|u\|$, never $\|v\|$ — measured along u

$|s|$ the shadow's length; $s > 0$ same side as u , $s < 0$ opposite, $s = 0$ perpendicular

$\varphi = 60^\circ$

acute: $s > 0$



Orthogonality and Orthonormal Bases

Predict the Sign

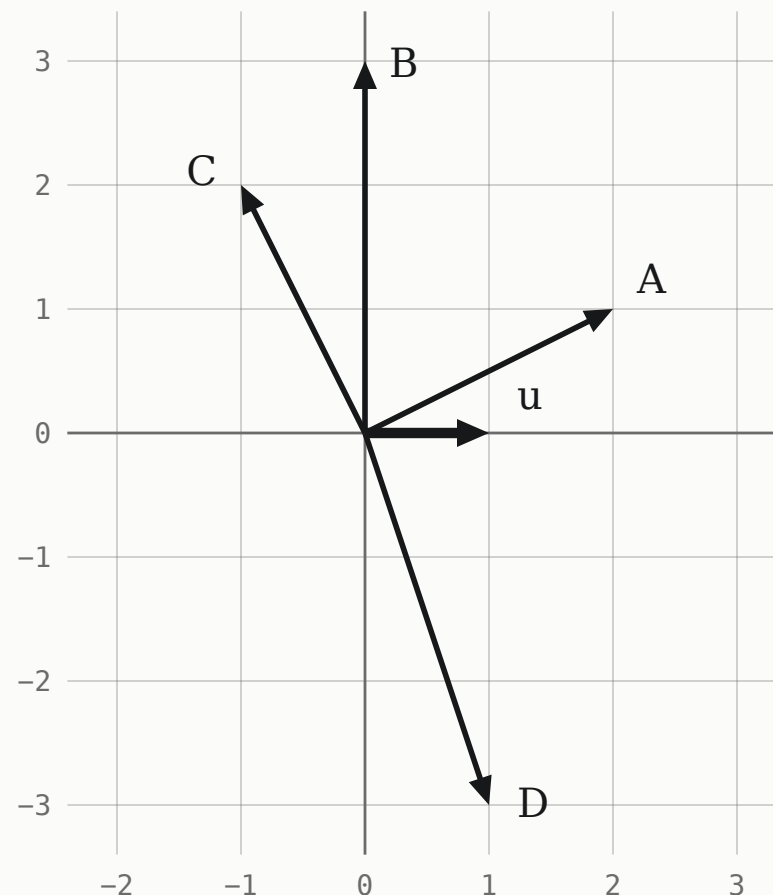
Which have a positive dot product with $u = (1, 0)$?

(A) $v_A = (2, 1)$

(B) $v_B = (0, 3)$

(C) $v_C = (-1, 2)$

(D) $v_D = (1, -3)$



Orthogonality and Orthonormal Bases

Orthogonality and the Pythagorean Theorem

$$u \perp v \iff u^T v = 0$$

$\perp$ orthogonal: dot product zero

$0^T v = 0$ so $0 \perp$ every v (no angle defined)

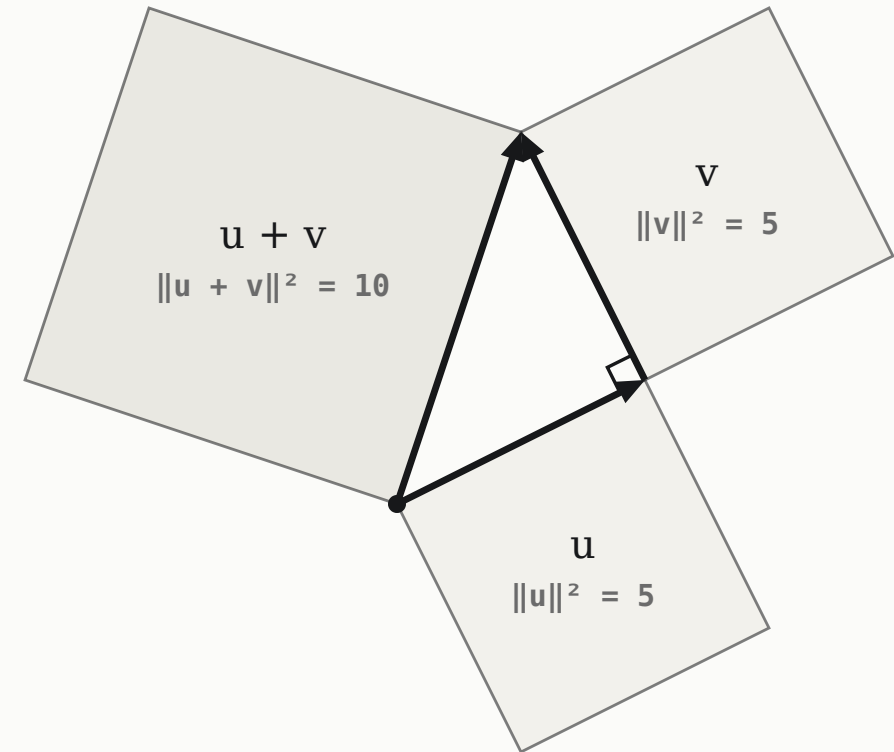
$$\|u + v\|^2 = \|u\|^2 + 2u^T v + \|v\|^2$$

why expand $(u + v)^T(u + v)$ by linearity

$2u^T v$ cross term: 0 iff $u \perp v$

WORKED EXAMPLE

$u = (2, 1)$, $v = (-1, 2)$: $u^T v = -2 + 2 = 0$, so $\|(1, 3)\|^2 = 10 = 5 + 0 + 5$.



DEFINITION

Orthogonality and Orthonormal Bases

Orthogonal to a Whole Subspace

$$r \perp S \iff r^T y = 0 \text{ for every } y \in S$$

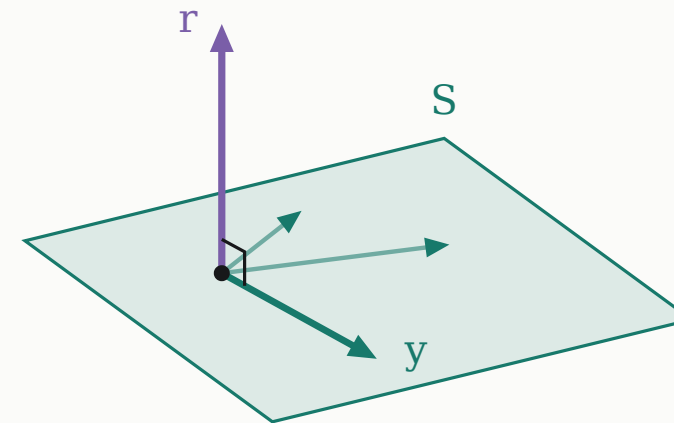
$S \subseteq \mathbb{R}^m$ a subspace; y is every vector of it

$r^T y$ one number for each y in S

$$S^\perp = \{r \in \mathbb{R}^m : r^T y = 0 \ \forall y \in S\}$$

a subspace $0 \in S^\perp$; $(c_1 r_1 + c_2 r_2)^T y = 0$ too, by linearity

edges $\{0\}^\perp = \mathbb{R}^m$, $(\mathbb{R}^m)^\perp = \{0\}$: $r^T r = 0 \Rightarrow r = 0$



schematic

Orthogonality and Orthonormal Bases

Perpendicular to a Line, or to the Plane?

$d_1 = (1, 0, 0)$, $d_2 = (0, 1, 0)$, $S = \text{span}\{d_1, d_2\}$, $r = (0, 1, 1)$

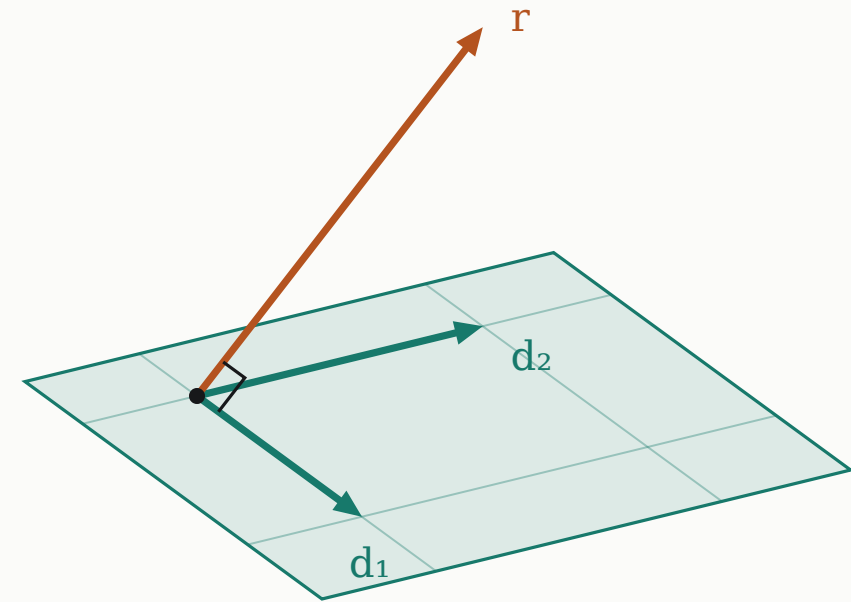
$r \perp d_1$. Which is true?

(A) $r \perp S$

(B) $r/\|r\| \perp S$

(C) $r \notin S$: $r^T d_2 = 1$

(D) $r^T d_1 = 0$ gives $r \perp S$



$S = \text{span}\{d_1, d_2\}$ 3D view: trust the number

Orthogonality and Orthonormal Bases

Testing a Spanning Set Is Enough

$$r^T y = \sum_{i=1}^k c_i (r^T d_i)$$

$y \in S$ $S = \text{span}\{d_1, \dots, d_k\}$, so $y = \sum c_i d_i$

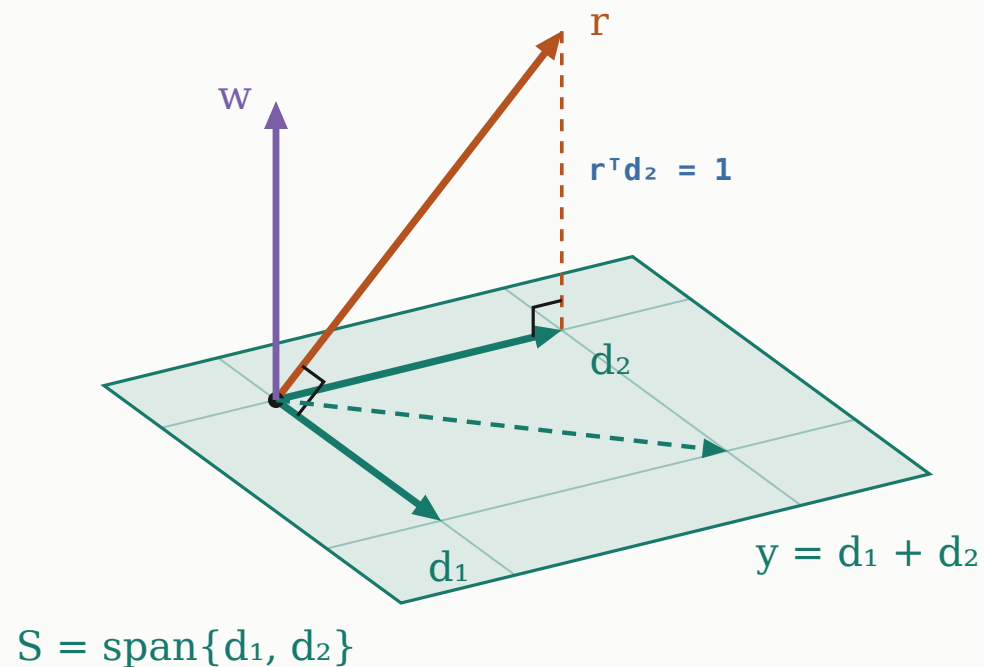
k tests settle every y : $r^T y$ is linear in y

span only independence not needed

WORKED EXAMPLE

From the vote: $y = d_1 + d_2$ gives $r^T y = 1$, $w^T y = 0$; d_1 alone misses it.

$r \perp S \Leftrightarrow r^T d_i = 0$ for every i .



Orthogonality and Orthonormal Bases

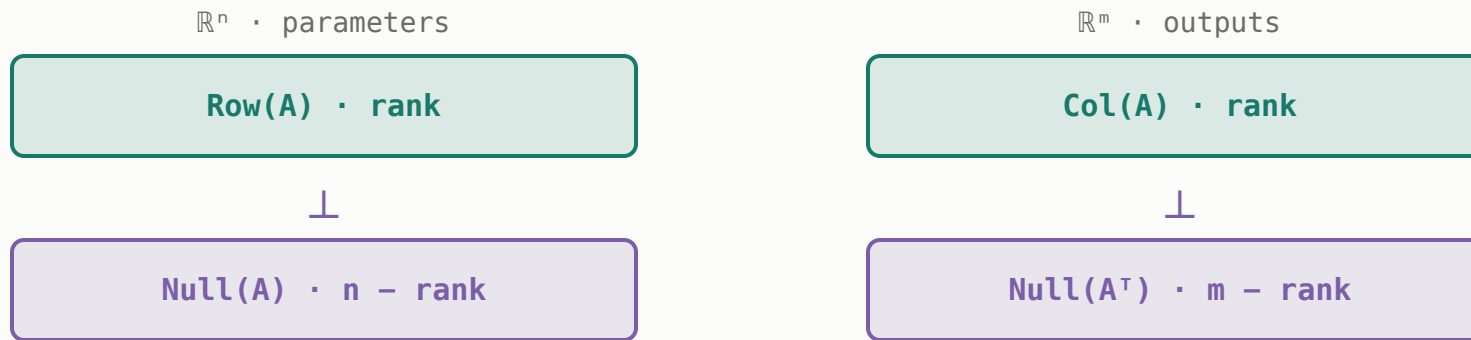
The Four Fundamental Subspaces Revisited

$A^T r = (a_1^T r, \dots, a_n^T r) \in \mathbb{R}^n$, so $r \perp \text{Col}(A) \Leftrightarrow A^T r = 0$.

$$\text{Col}(A)^\perp = \text{Null}(A^T) \subseteq \mathbb{R}^m, \quad \text{Null}(A) = \text{Row}(A)^\perp \subseteq \mathbb{R}^n$$

a_j columns of $A \in \mathbb{R}^{m \times n}$: they span $\text{Col}(A)$

input same on A^T ; Lecture 1 wrote $\mathcal{C}, \mathcal{N}$



Strang, G. **MIT 18.06 Linear Algebra**, Lecture 14, orthogonal vectors and subspaces. MIT OpenCourseWare, 2010.

Orthogonality and Orthonormal Bases

Orthogonal Is Not the Same as Orthonormal

$$q_i^\top q_j = \delta_{ij}$$

δ_{ij} 1 if $i = j$ (unit), else 0
(orthogonal)

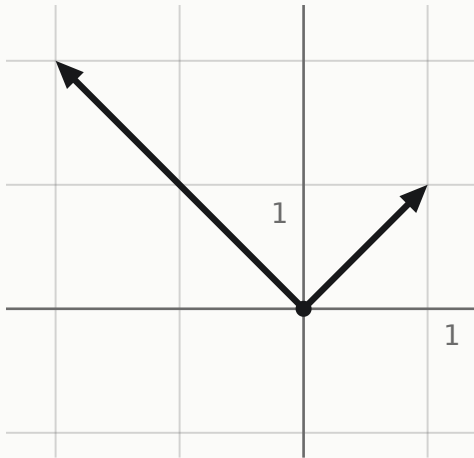
Each pair is a basis. Sort:

(A) independent only

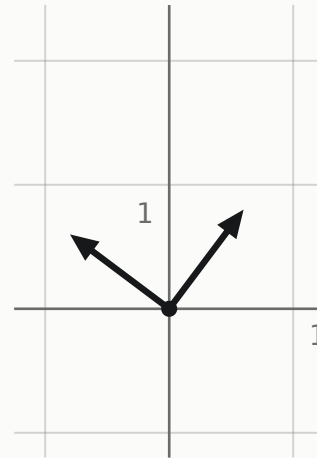
(B) orthogonal, not unit

(C) orthonormal

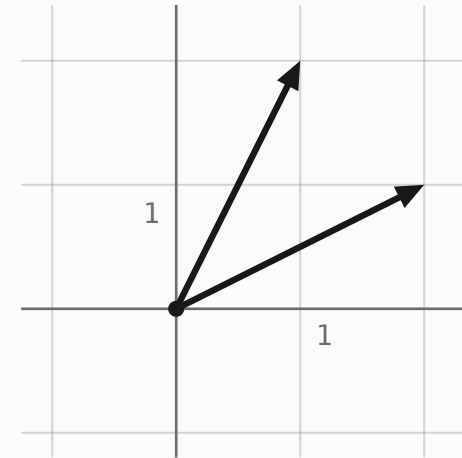
1



2



3



Orthogonality and Orthonormal Bases

Right Angles, Unit Lengths, Independence

RIGHT ANGLES

$$\sum_i c_i v_i = 0 \Rightarrow v_j^T \sum_i c_i v_i = c_j \|v_j\|^2 = 0$$

v_i pairwise $\perp$, nonzero: $\|v_j\|^2 > 0$

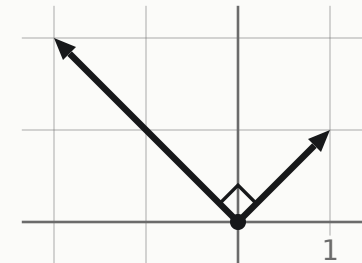
all j $c_j = 0$: independent

fails drop nonzero: $0 \perp$ all

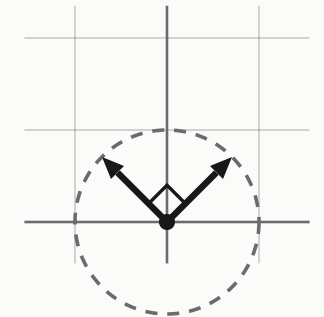
UNIT LENGTHS

Not for independence; with $\perp$: $v = \sum c_i v_i \Rightarrow c_j = v_j^T v$.

Orthonormal = both. Nonzero right angles $\Rightarrow$ independent, not conversely.



orthogonal



orthonormal

Orthogonality and Orthonormal Bases

What Does $Q^T Q = I$ Actually Mean?

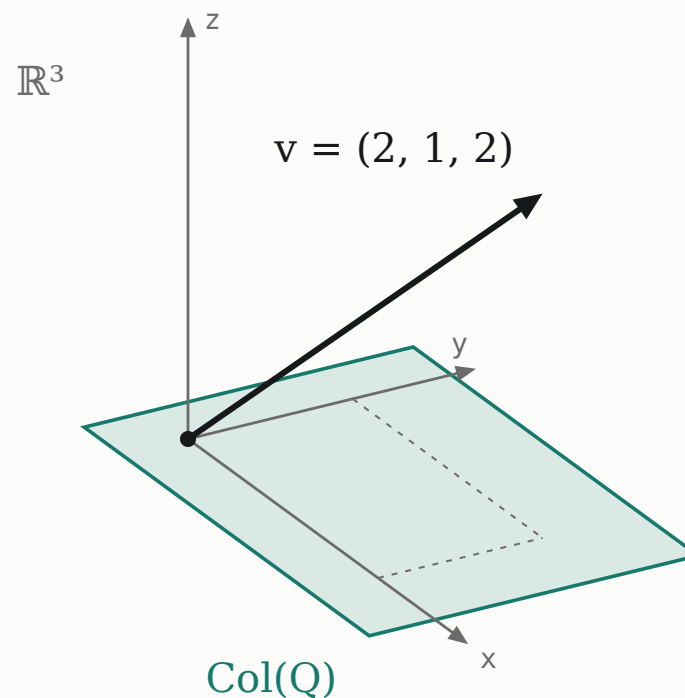
For $Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$, which are true?

(A) $Q^T Q = I_2$

(B) $Q Q^T = I_3$

(C) $\|Qz\| = \|z\|$ for every $z \in \mathbb{R}^2$

(D) $Q^T v$ keeps all of any $v \in \mathbb{R}^3$



Orthogonality and Orthonormal Bases

Q Embeds, Q^T Extracts

Q EMBEDS

$$\|Qz\|^2 = z^T Q^T Q z = \|z\|^2$$

Q $m \times k$, columns q_i , $Q^T Q = I_k$; $S = \text{Col}(Q)$

$z \in \mathbb{R}^k$ $Qz \in \mathbb{R}^m$: lengths, angles kept

Q^T EXTRACTS

$(Q^T b)_i = q_i^T b$: coordinates of the projection.

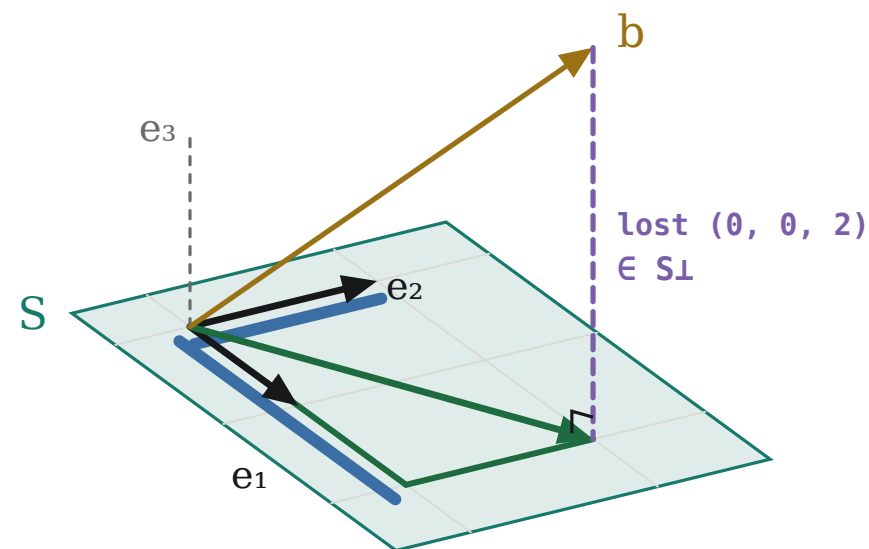
	Full orthonormal basis	Orthonormal basis of a proper subspace
$Q^T Q$	I_m	I_k
$Q Q^T b$	b	projection
recover b yes		iff $b \in S$

Q^T loses b 's $S^\perp$ component.

$$Q = [e_1 \ e_2] \quad k = 2$$

$$Q^T b = (2, 1)$$

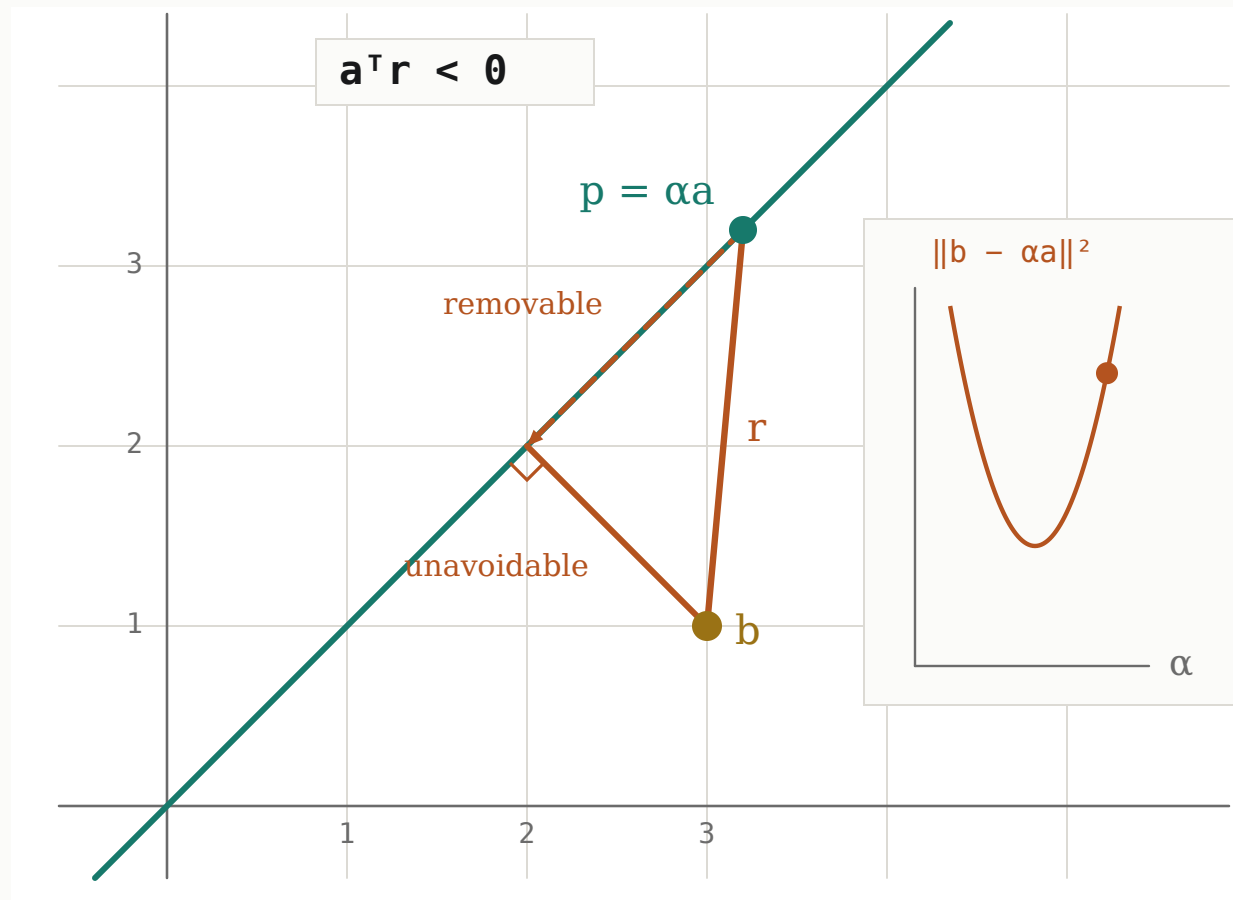
$$Q Q^T b = (2, 1, 0)$$



$$\mathbf{a} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

Orthogonal Projection

The Residual Must Be Perpendicular



$$\mathbf{r} = \mathbf{b} - \mathbf{p}, \quad \mathbf{p} = \alpha \mathbf{a}$$

- $\mathbf{b}$ the target in $\mathbb{R}^m$, off the line
- $\mathbf{p} = \alpha \mathbf{a}$ a candidate; α is our only choice
- $\mathbf{r}$ what $\mathbf{p}$ fails to explain

$$\text{lowest } \|\mathbf{b} - \alpha \mathbf{a}\|^2 \Rightarrow \mathbf{a}^T \mathbf{r} = 0$$

- $\mathbf{a}^T \mathbf{r} \neq 0$ $\mathbf{r}$ keeps a part along the line: removable, so $\mathbf{p}$ is not lowest
- $\mathbf{a}^T \mathbf{r} = 0$ nothing left along the line — why is that enough?

DEFINITION

Orthogonal Projection

Solving the Perpendicularity Equation

$a \neq 0, b \in \mathbb{R}^m$. Candidate on $\text{span}\{a\}$: $p = \alpha a$

Ask for a perpendicular residual: $a^\top(b - \alpha a) = 0$

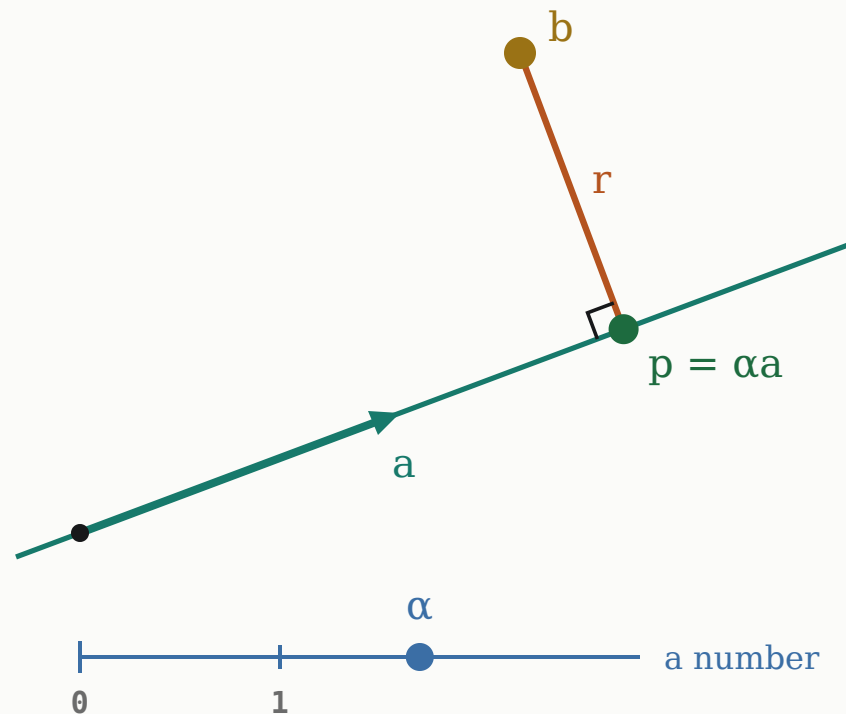
Expand: $a^\top b - \alpha a^\top a = 0$ — one equation, one unknown

$$\alpha = \frac{a^\top b}{a^\top a}, \quad p = \alpha a$$

α a number: how many copies of a

p a vector in $\mathbb{R}^m$

$a^\top a = \|a\|^2 > 0$ as $a \neq 0$; $a = 0$: no line



Strang. **MIT 18.06**, L15 projections onto subspaces. MIT OpenCourseWare, 2010.

Orthogonal Projection

The Same Projection as a Matrix

Substitute α back: the same p , now one map.

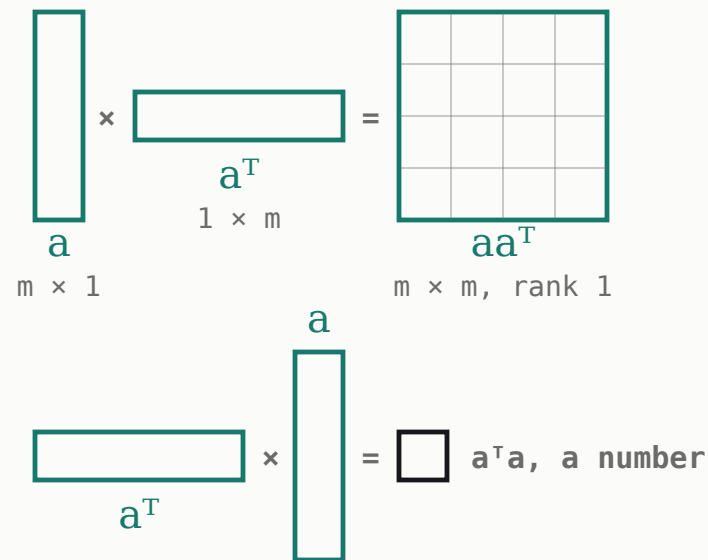
$$p = a \frac{a^\top b}{a^\top a} = \frac{aa^\top}{a^\top a} b$$

$a^\top b$ $1 \times m$ by $m \times 1$: a number

$aa^\top$ $m \times 1$ by $1 \times m$: $m \times m$, rank 1 if $a \neq 0$, symmetric

$a = 0$ divisor 0: quotient undefined

Types: α number $\cdot$ p vector $\cdot$ $\frac{aa^\top}{a^\top a}$ matrix



$$\mathbf{a} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

Orthogonal Projection

Compute and Check the Projection

Alone, no arithmetic: which way does r point from p to b , and is α above or below 1?

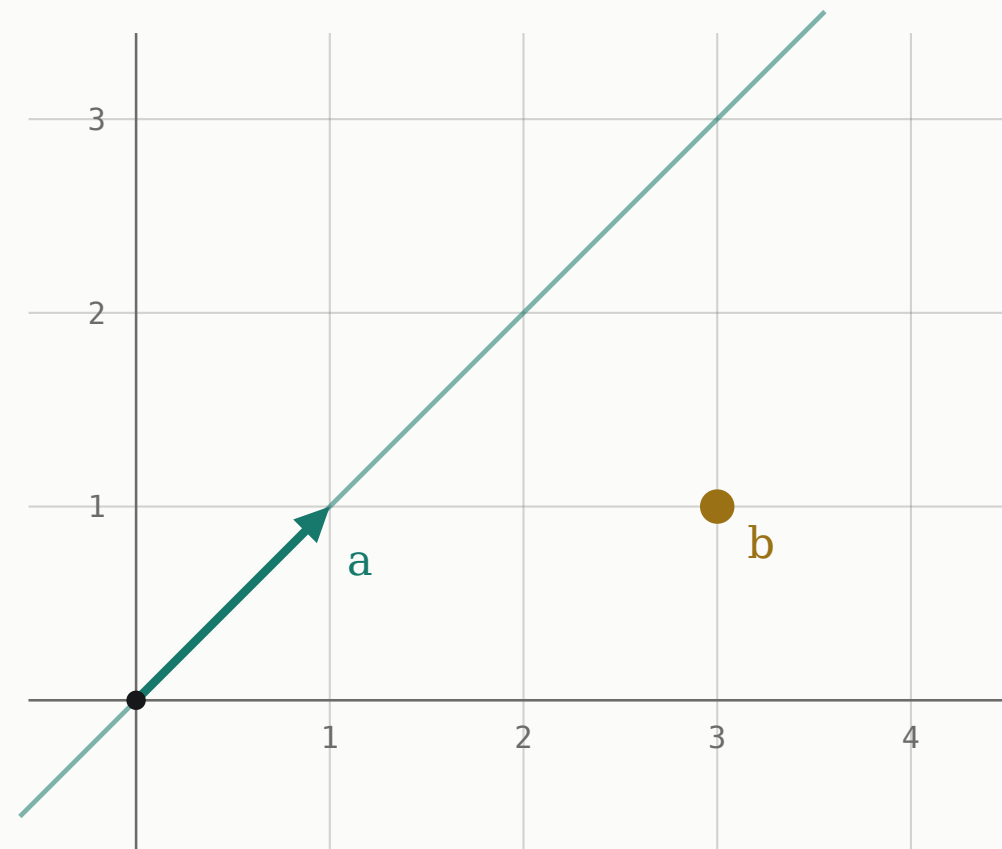
Then in pairs: α , p , r , check r .

1 α

2 p

3 r

CHECK

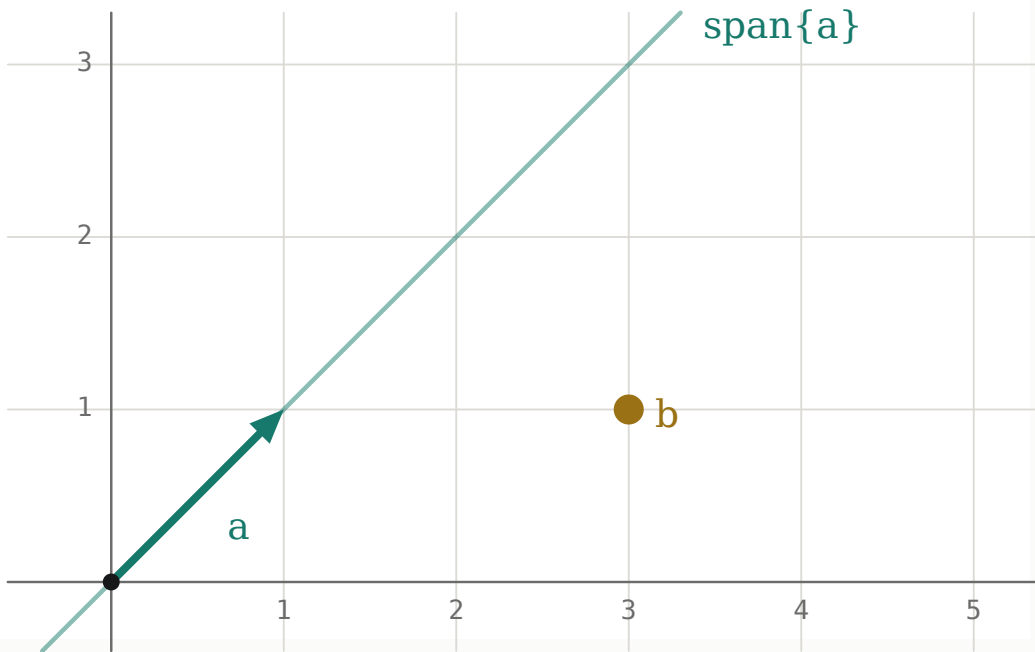


$$\mathbf{a} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

Orthogonal Projection

Does Rescaling the Basis Change the Projection?

$$\lambda = 1.00$$



Replace $\mathbf{a}$ by $\lambda\mathbf{a}$, $\lambda > 0$. What happens to $\mathbf{a}^\top\mathbf{b}$, α , p ?

- (A) $\mathbf{a}^\top\mathbf{b} \times \lambda$, $\alpha \div \lambda$, p same
- (B) all the same
- (C) $\mathbf{a}^\top\mathbf{b} \times \lambda$, $\alpha \times \lambda$, $p \times \lambda^2$
- (D) $\mathbf{a}^\top\mathbf{b}$ same, $\alpha \div \lambda$, p same

Orthogonal Projection

Component, Coefficient, Projection

$a \neq 0$

$\lambda a, \lambda > 0$

$$s = a^\top b / \|a\|$$

signed shadow

same

$$\alpha = a^\top b / \|a\|^2$$

copies of a

$\div \lambda$

$$p = \alpha a$$

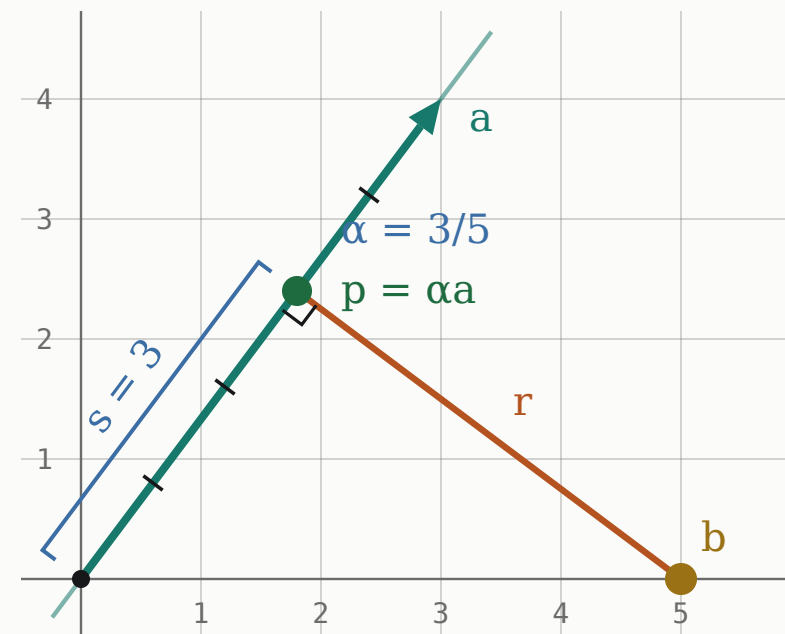
vector in $\mathbb{R}^m$

same

Unit a : $s = \alpha$, the coordinate of Lecture 2; else equal only if $a^\top b = 0$.

WORKED EXAMPLE

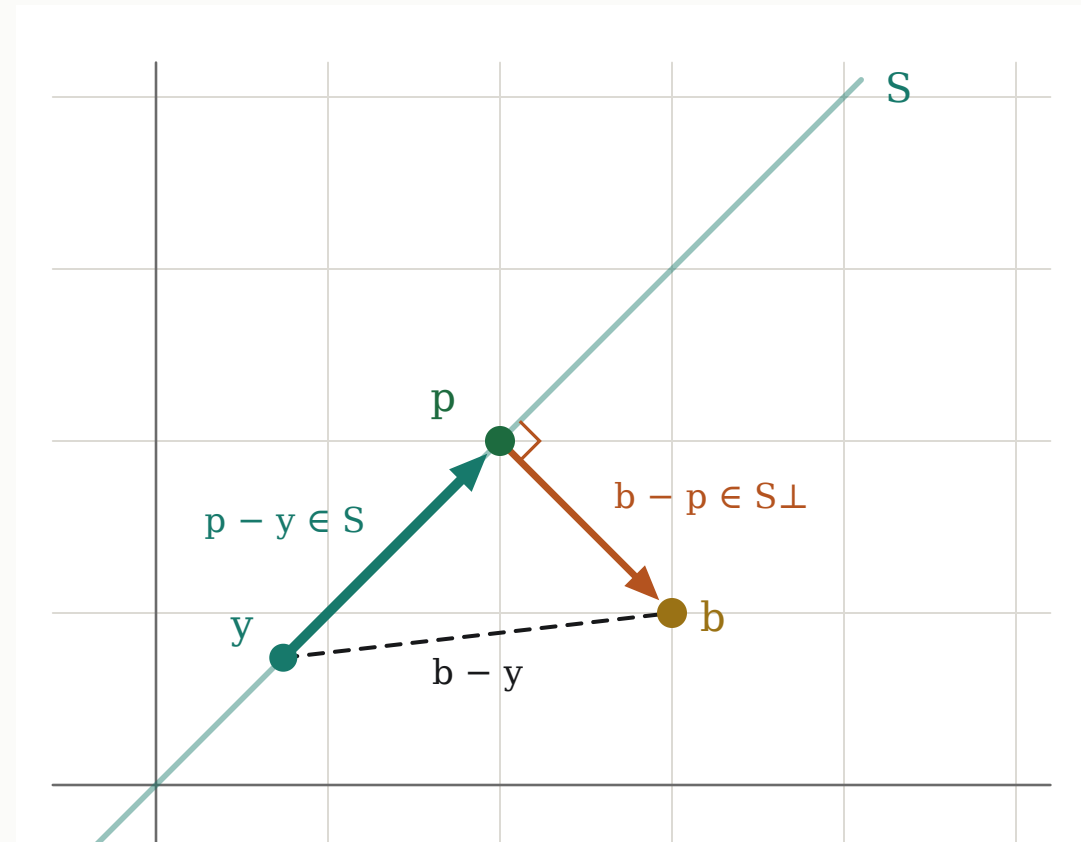
$a = (3, 4), b = (5, 0): a^\top b = 15, \|a\| = 5, s = 3, \alpha = 3/5,$
 $p = (9/5, 12/5), r = (16/5, -12/5), a^\top r = 0$



$$\mathbf{a} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

Orthogonal Projection

Split the Error at Any Candidate



Why is a perpendicular residual enough to make p the closest point?

Premise: S any subspace, $p \in S$, $r = b - p \in S^\perp$. Take any other $y \in S$.

$$b - y = (b - p) + (p - y)$$

$p - y$ lies in S : a subspace is closed under subtraction

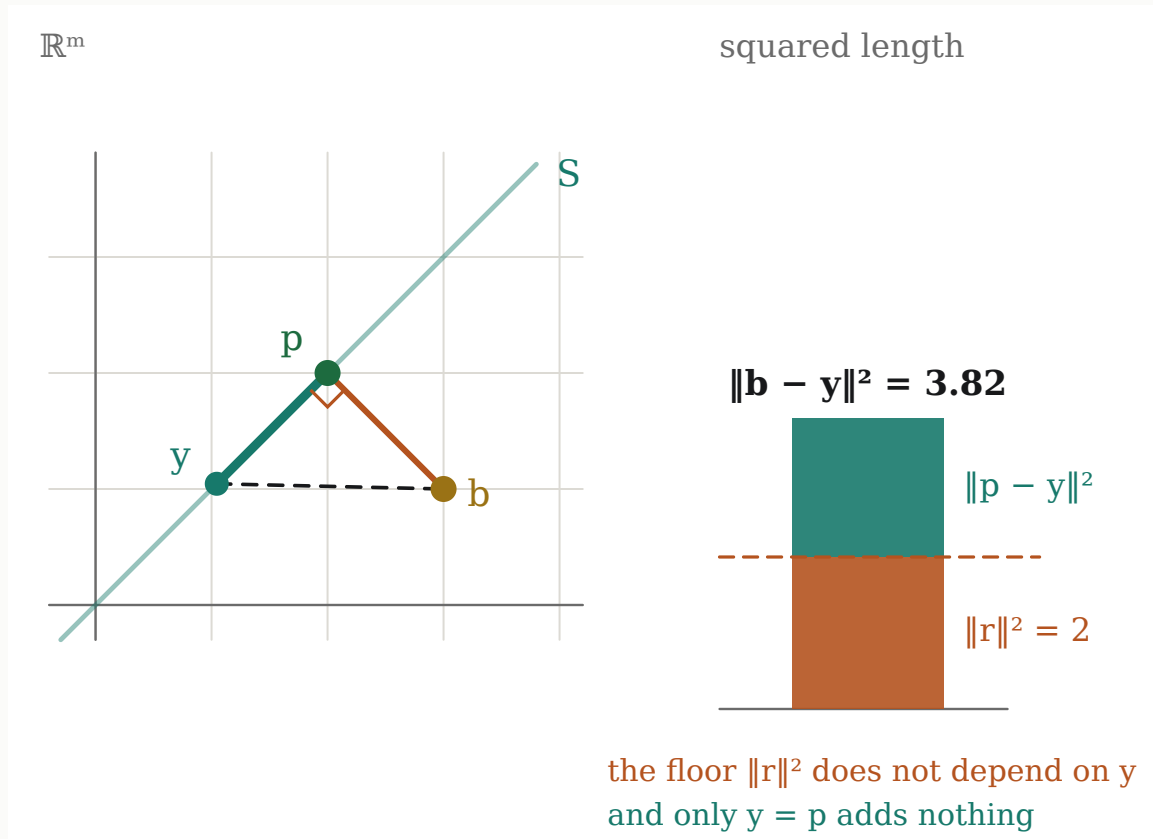
$b - p$ lies in $S^\perp$, so $(b - p)^\top (p - y) = 0$

right angle the two pieces are the legs, $b - y$ the hypotenuse

$$a = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad b = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

Orthogonal Projection

Yes: the Perpendicular Test Is Enough



Yes. Take squared lengths on that right triangle.

$$\|b - y\|^2 = \|r\|^2 + \|p - y\|^2 \geq \|r\|^2$$

$\|r\|^2$ unavoidable: it does not depend on y

$\|p - y\|^2$ the extra, ≥ 0 , zero only at $y = p$

Optimal · every $y \neq p$ adds a positive square

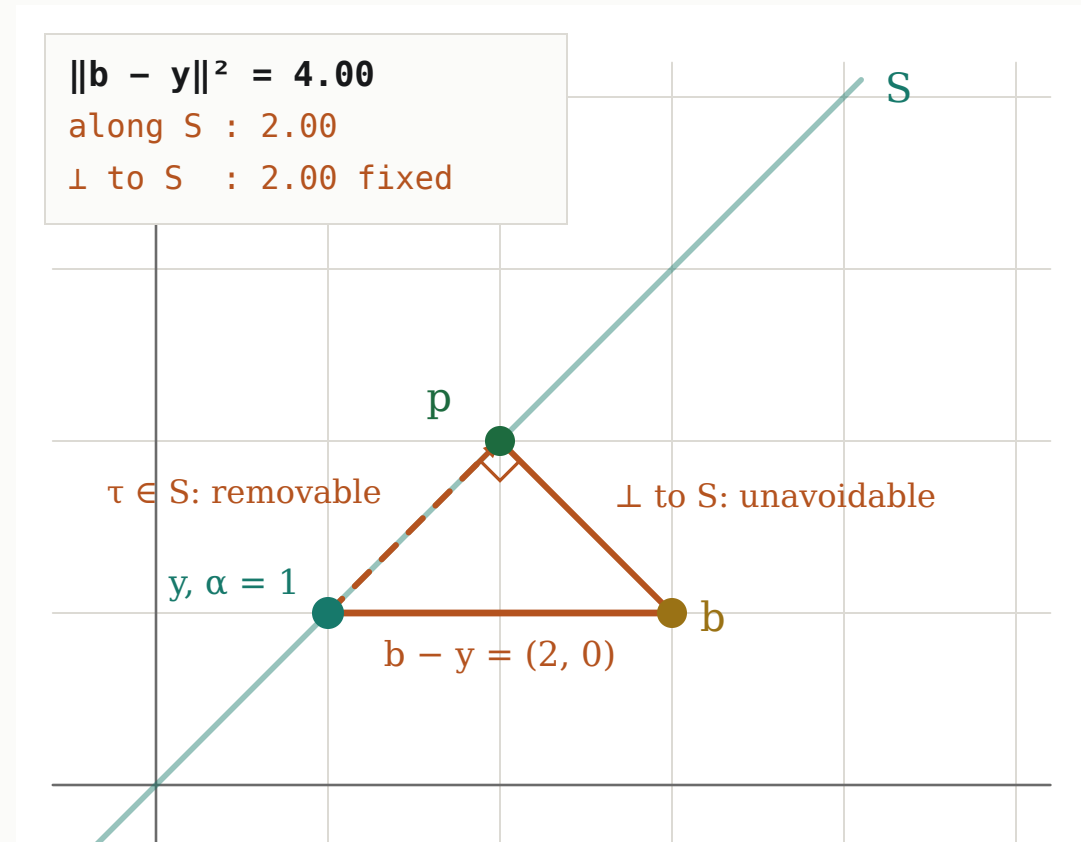
Unique · nothing else ties p

Built for a line; a general S still needs a construction.

$$a = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad b = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

Orthogonal Projection

A Candidate That Is Not Yet Best



An illustration of the converse: a residual still leaning along S .

$$\|(b - y) - \tau\|^2 = \|b - y\|^2 - \|\tau\|^2$$

τ the orthogonal projection of $b - y$ onto the line;
 step to $y + \tau \in S$

$b - y - \tau$ the perpendicular part, untouched

why Pythagoras: $\tau \perp (b - y - \tau)$

At $\alpha = 1$: $b - y = (2, 0)$, $\tau = (1, 1)$; at $y + \tau = p$: $4 - 2 = 2$,
 nothing left to improve.

DEFINITION

Orthogonal Projection

The Projection Theorem

$S \subseteq \mathbb{R}^m$ a subspace: each $b \in \mathbb{R}^m$ splits one way only, and that p is closest.

$$b = p + r, \quad p \in S, \quad r \in S^\perp$$

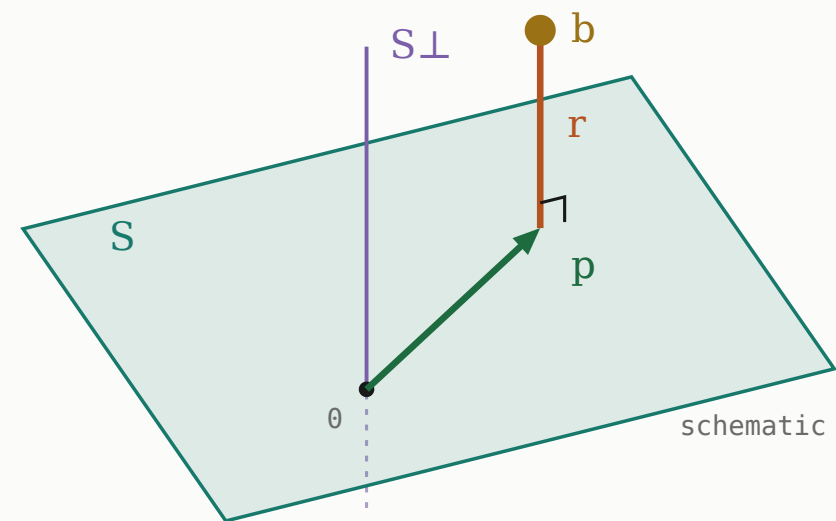
unique $p + r = p' + r' \Rightarrow p - p' = r' - r \in S \cap S^\perp = \{0\}$ ($\perp$ itself)

closest proved, for any $y \in S$

exists construction follows, from an orthonormal basis Q

edges $b \in S \Rightarrow r = 0; b \in S^\perp \Rightarrow p = 0$

Lecture 1 $S = \text{Col}(A)$: $b = \hat{b} + r$, $r \in \text{Null}(A^\top)$, dims add



Orthogonal Projection

Projection onto a Subspace: $p = QQ^T b$

$$Q = \begin{bmatrix} 1 & 0 \\ 0 & 4/5 \\ 0 & 3/5 \end{bmatrix} \quad b = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$$

$Q \in \mathbb{R}^{m \times k}$ with $Q^T Q = I_k$: columns q_i orthonormal, spanning S .

$$c = Q^T b \in \mathbb{R}^k, \quad p = Qc = QQ^T b$$

extract $c_i = q_i^T b$: the part of b along q_i

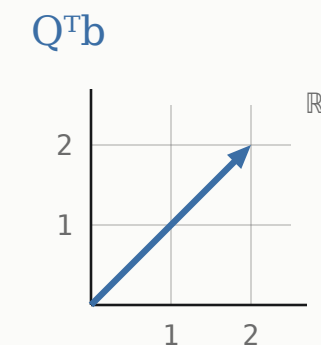
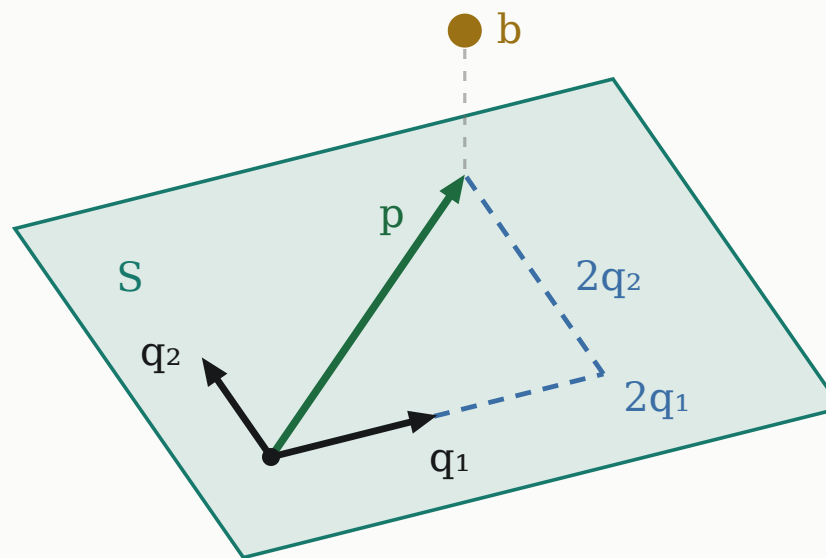
rebuild $p = c_1 q_1 + \dots + c_k q_k \in S$

c the coordinates of p ; of b too iff $b \in S$

sizes $m \times k$ times $\mathbb{R}^k$: $p \in \mathbb{R}^m$

WORKED EXAMPLE

$$Q^T b = (2, 2), \quad p = (2, 8/5, 6/5)$$



Orthogonal Projection

Checking the Two Requirements

$$Q = \begin{bmatrix} 1 & 0 \\ 0 & 4/5 \\ 0 & 3/5 \end{bmatrix} \quad b = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$$

Both requirements, one at a time.

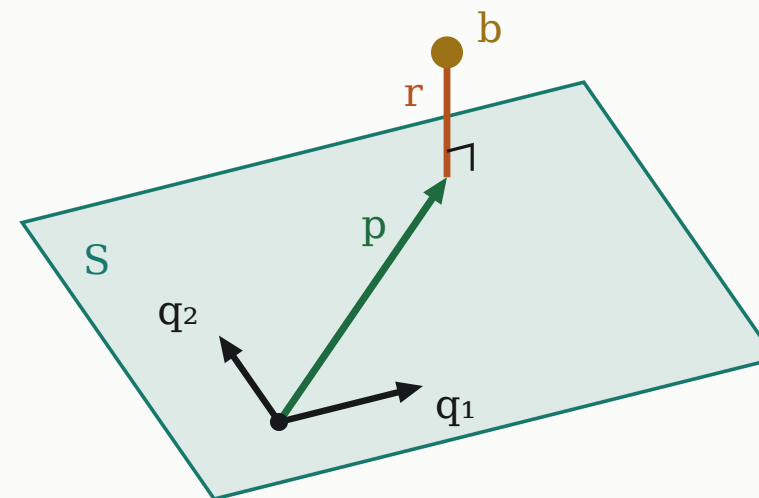
1. $p \in S$: $p = Q(Q^T b)$, Q times a vector of $\mathbb{R}^k$.

$$Q^T(b - p) = Q^T b - (Q^T Q)Q^T b = 0$$

2. $b - p \perp$ each $q_i \Rightarrow \perp S$ (spanning test) $\Rightarrow p$ is **the** closest point

needs $Q^T Q = I_k$; a general basis A needs a correction factor, not plain AA^T

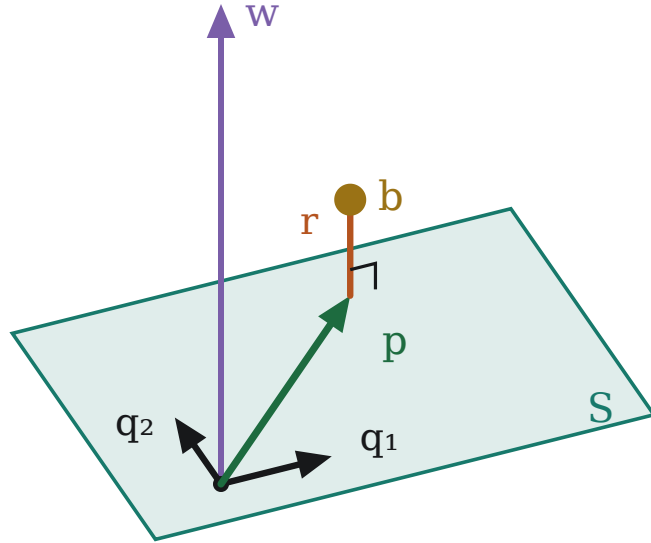
here $r = (0, -3/5, 4/5)$: $q_1^T r = 0$, $q_2^T r = -12/25 + 12/25 = 0$



$$Q = \begin{bmatrix} 1 & 0 \\ 0 & 4/5 \\ 0 & 3/5 \end{bmatrix} \quad b = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$$

Orthogonal Projection

Does Moving b Out of the Plane Change the Prediction?



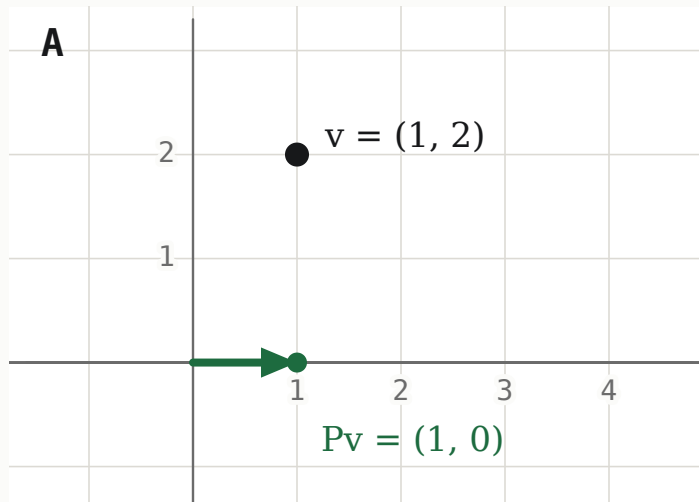
Same S and b . Add $w = (0, -3, 4)$, $\perp q_1$ and q_2 . For $b + w = (2, -2, 6)$:

- ☐ (A) projection changes, residual same
- ☐ (B) projection same, residual changes
- ☐ (C) both stay the same
- ☐ (D) both change

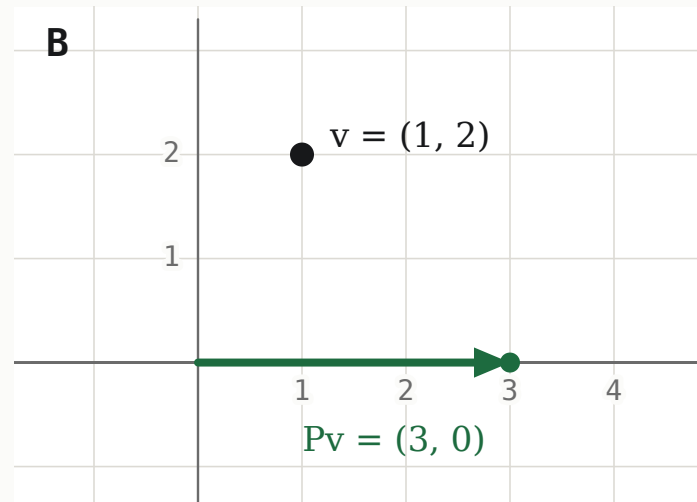
Orthogonal Projection

Which Matrix Is an Orthogonal Projector?

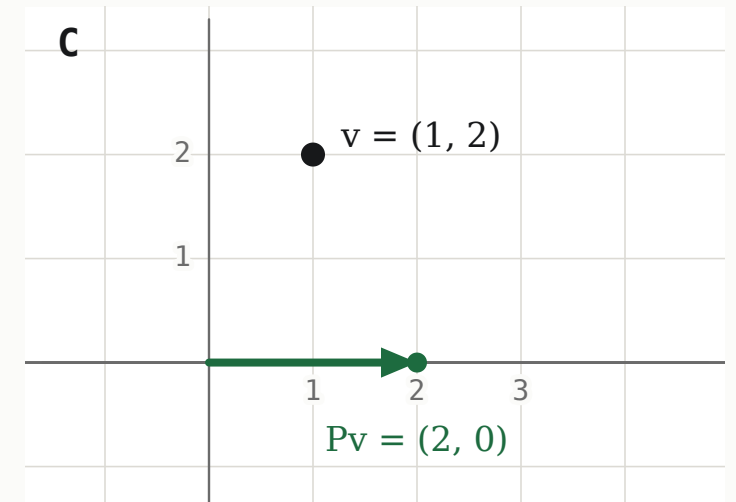
Each of these flattens the plane onto the axis. Which sends every v to its orthogonal projection?
On $v = (1, 2)$: does a second pass move Pv ? Is the discarded direction $\perp$ the axis?



A $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$



B $\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$



C $\begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$

DEFINITION

Orthogonal Projection

What the Projection Matrix Does

Orthogonal projector onto S : a matrix P with $Pb \in S$ and $b - Pb \in S^\perp$ for **every** $b \in \mathbb{R}^m$ (the two requirements checked).

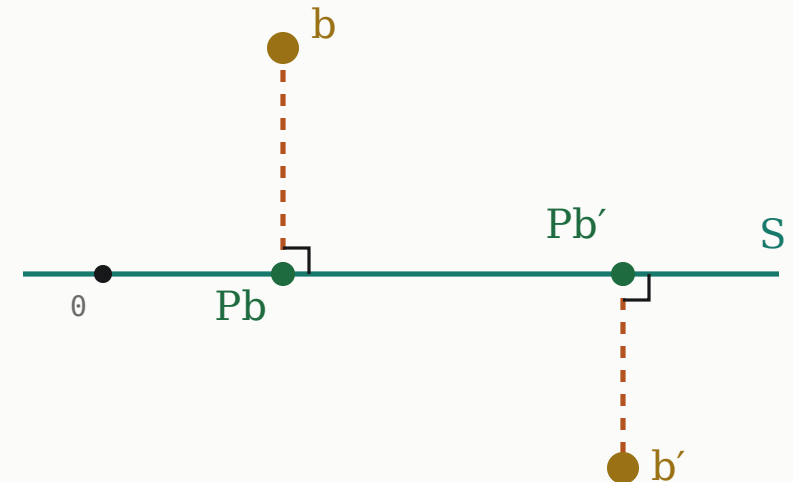
$$P = QQ^\top \in \mathbb{R}^{m \times m}, \quad Pb = p$$

why $QQ^\top$ passes both; the closest point is unique

size Q is $m \times k$; P is $m \times m$, rank k , symmetric

unique any orthonormal basis of S gives the same P

edges $S = \{0\}$: $P = 0$; $S = \mathbb{R}^m$: $P = I$



Orthogonal Projection

Applying It Twice Changes Nothing

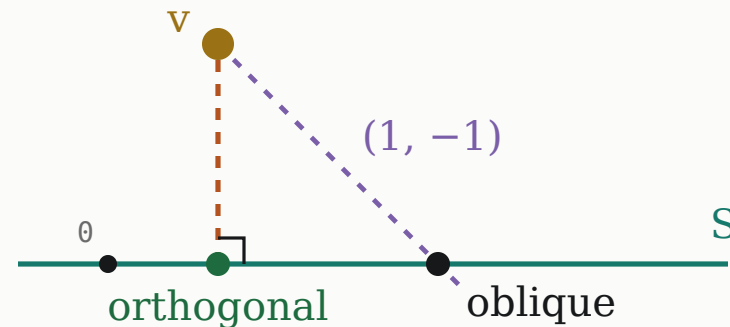
$$P^2 = Q(Q^\top Q)Q^\top = QQ^\top = P$$

reason $Pb \in S$, and the closest point of S to a point of S is itself

reads $P(Pb) = Pb$ for every b : the second pass moves nothing

uses $Q^\top Q = I_k$, nothing else

$P^2 = P$ alone means **a** projection, not necessarily orthogonal: the vote's P_B is idempotent, yet $(1, -1)$, the direction it discards, is not $\perp$ the axis.



Orthogonal Projection

Symmetry Is the Missing Half

$$P^T = (QQ^T)^T = QQ^T = P$$

why $(UV)^T = V^T U^T$, $(Q^T)^T = Q$
so far QQ^T : symmetric, idempotent

$P^T = P = P^2 \Leftrightarrow$ orthogonal projector onto $\text{Col}(P)$

$(\Rightarrow) Pb \in \text{Col}(P); P^T(b - Pb) = P(b - Pb) = Pb - P^2b = 0: b - Pb \in \text{Col}(P)^\perp$

$(\Leftarrow) P$ fixes $\text{Col}(P)$: $P^2 = P$. Any a, b : $a^T Pb = (Pa)^T Pb$ [$a - Pa \perp Pb$] $= (Pa)^T b$ [$b - Pb \perp Pa$] $= a^T P^T b$: $P^T = P$

vote: $P_B \neq P_B^T$, $P_C^2 \neq P_C$

Orthogonal Projection

What P Keeps and What It Discards

$$(I - P)^T = I - P, \quad (I - P)^2 = I - 2P + P^2 = I - P$$

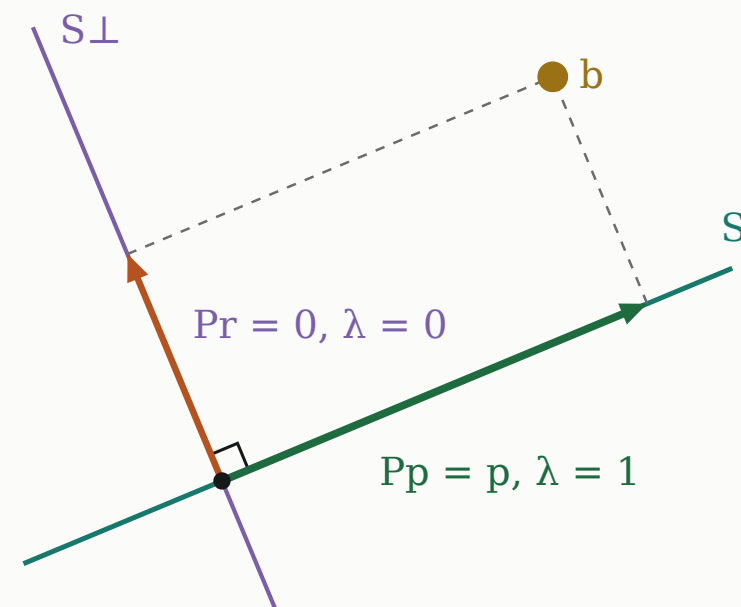
$I - P$ orthogonal projector onto $S^\perp$: $Q^T(I - P)b = 0$, $P(I - P) = 0$

$b = p + r$ $Pp = p$, $Pr = 0$

$\lambda = 1$ on kept $p \neq 0$, 0 on discarded $r \neq 0$ (Lecture 2)

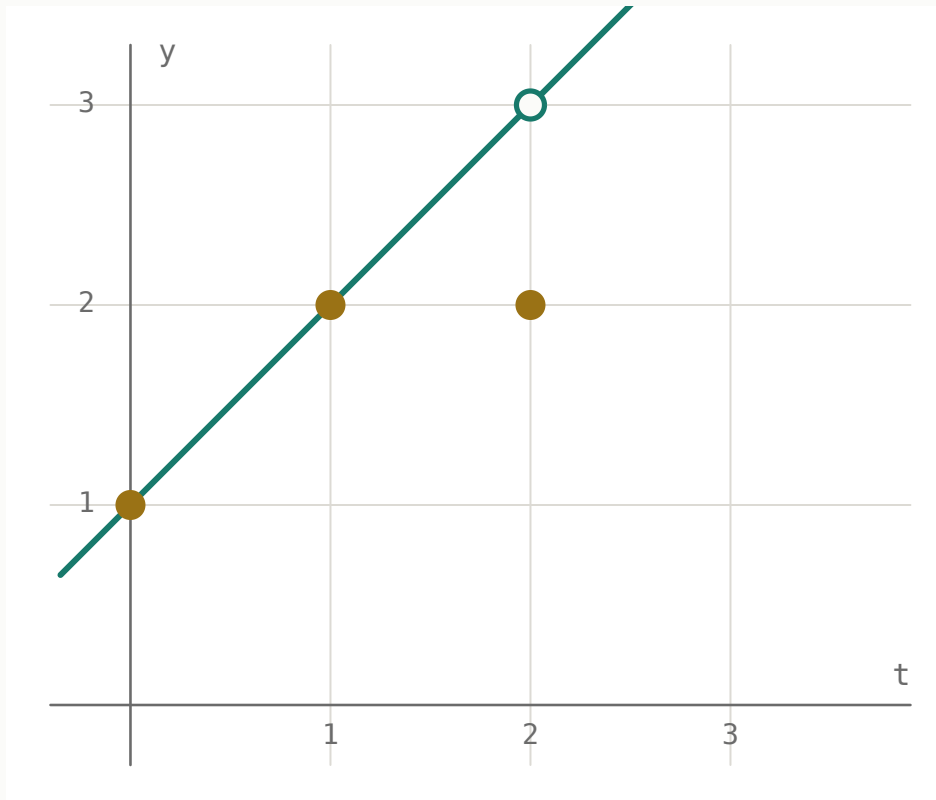
$\lambda = 0$ P singular: $P(b + w) = Pb$ for $w \in S^\perp$, so Pb does not determine the $S^\perp$ part of b

edges $P = 0$: no $\lambda = 1$; $P = I$: no $\lambda = 0$



Least Squares

Three Measurements, Two Unknowns



Model $y \approx \beta_0 + \beta_1 t$ at inputs t_i .

$$\begin{bmatrix} \boxed{1} & \boxed{0} \\ \boxed{1} & \boxed{1} \\ \boxed{1} & \boxed{2} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} \approx \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

all ones $\cdot \beta_0$, the shared height
the inputs $t_i \cdot \beta_1$, the slope

$A \in \mathbb{R}^{3 \times 2}$ $m = 3$ readings, $n = 2$ parameters

$x = (\beta_0, \beta_1)$ intercept and slope, the unknowns

b the readings y_i , one per input t_i

$b \notin \text{Col}(A)$ no exact x

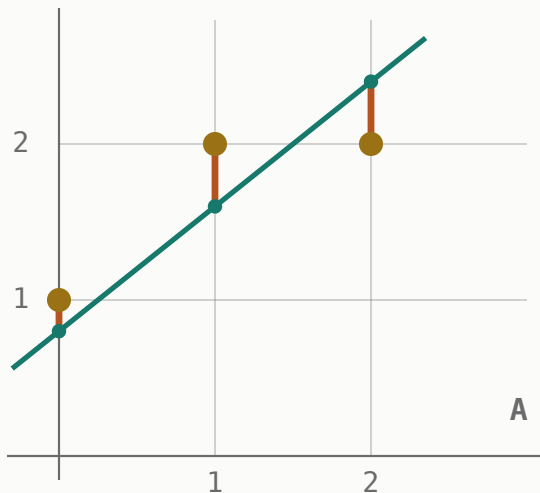
VOTE

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

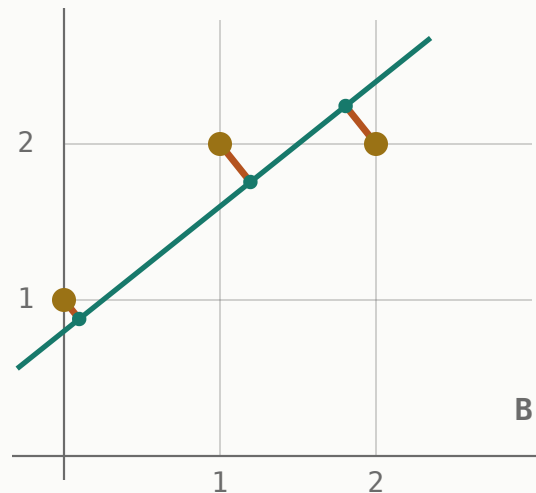
Least Squares

Which Errors Are We Minimizing?

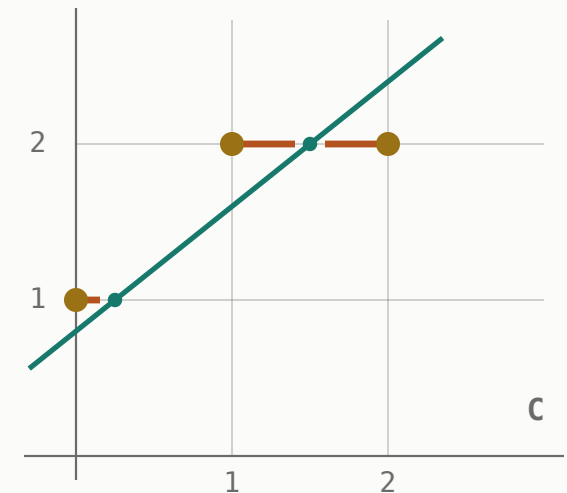
For $y_i \approx \beta_0 + \beta_1 t_i$, which errors does ordinary least squares minimize?



(A) vertical distances, squared



(B) shortest distances, squared

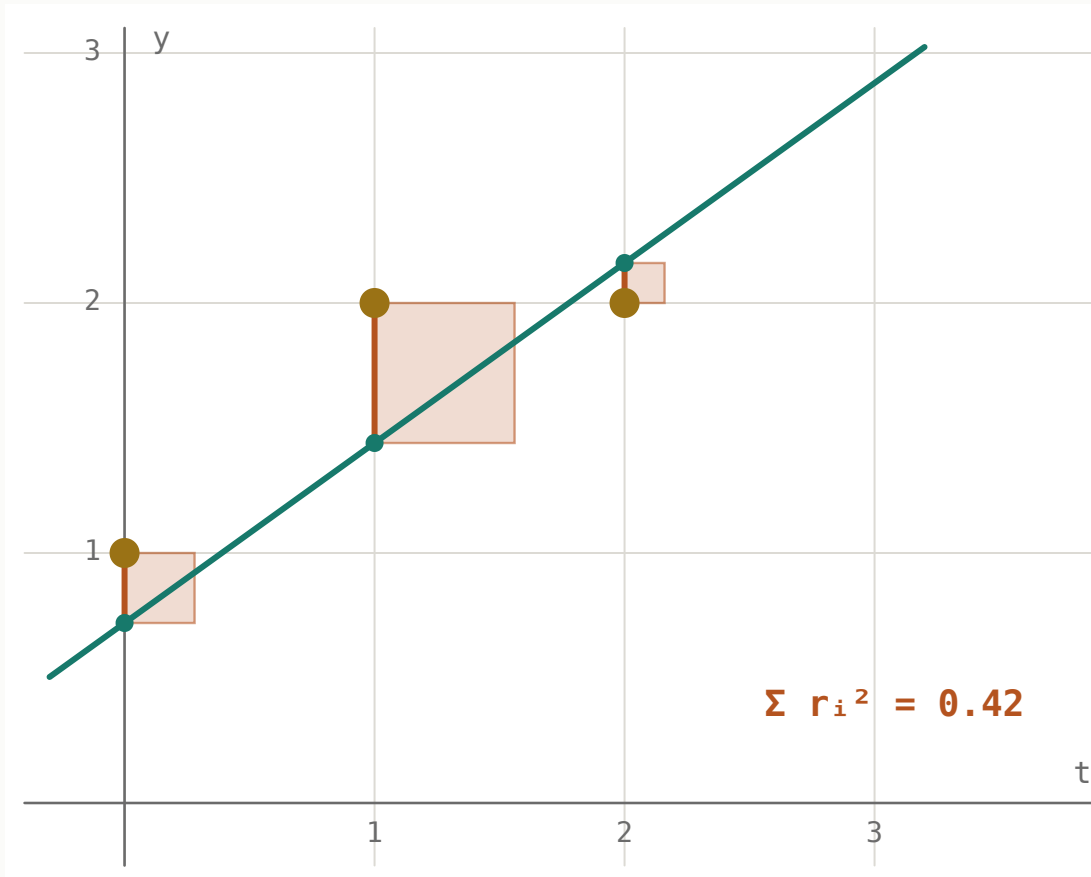


(C) horizontal distances, squared

Least Squares

The Objective and Its Objects

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$



$$r = b - Ax, \quad f(x) = \|Ax - b\|_2^2 = \|r\|_2^2$$

$x \in \mathbb{R}^n$ the parameters (β_0, β_1) we choose

$Ax, b, r \in \mathbb{R}^m$ prediction, readings, residual:
measurement space

$f(x)$ one number, ≥ 0 ; any A , no rank
assumption

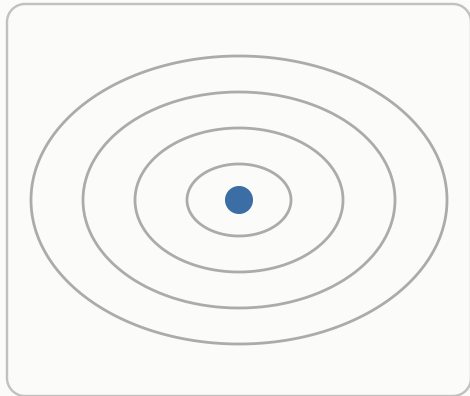
$|r_i|$ the side of square i — a side cannot be
negative

$e = -r$ Lecture 5 writes $e = Ax - b$

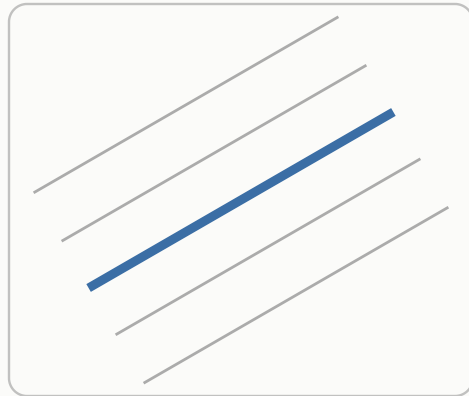
$cf, c > 0$ same minimiser, gradient $\times c$

Least Squares

Minimum and Minimiser



one bottom



a flat valley

both in parameter space $\mathbb{R}^n$ schematic

$$\min_x f(x) \in \mathbb{R}, \quad \hat{x} \in \arg \min_x f(x) \subseteq \mathbb{R}^n$$

min the smallest value f reaches — one number
(ours: p53)

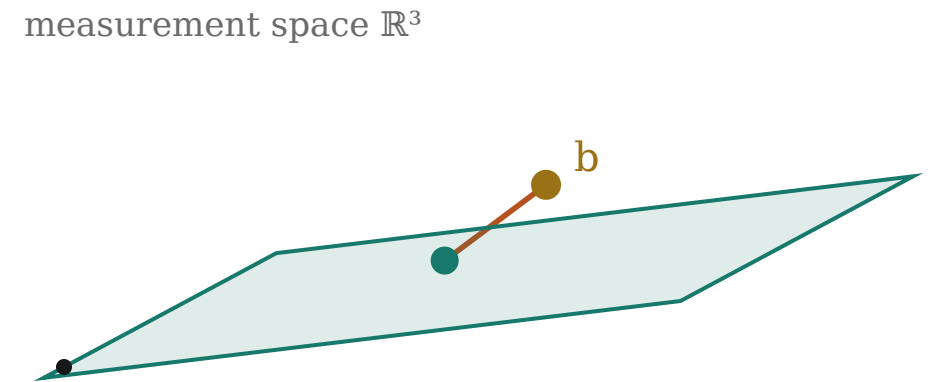
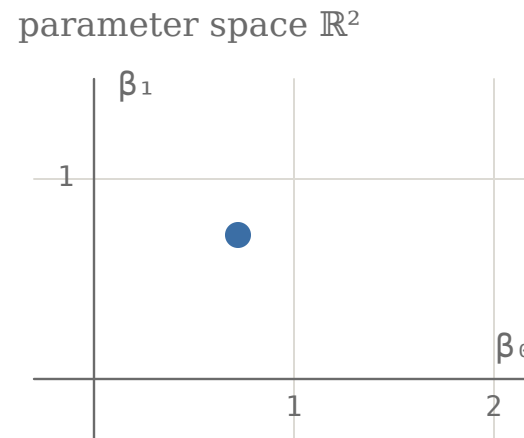
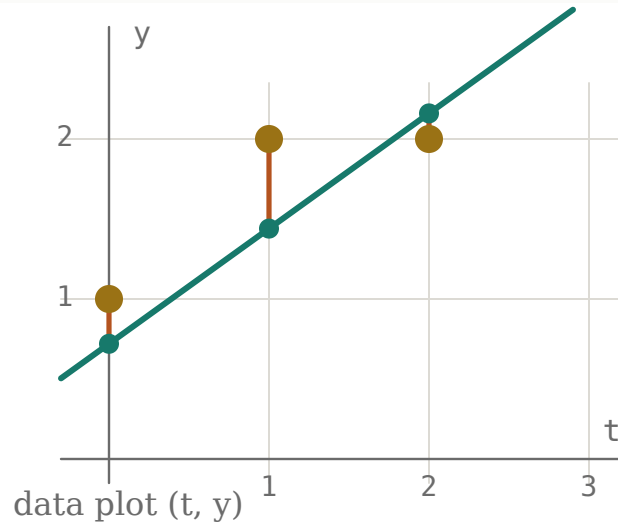
arg min the set of x reaching it: one point or many
 $\hat{x}$ any member of that set; any A , any rank

The best output $A\hat{x}$ is always unique; the best parameter vector $\hat{x}$ need not be.

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

Least Squares

Fitting Is Projection in Measurement Space



What kind of object is each, a line, a point or a plane, and in which space?

(A) the numbers (β_0, β_1)

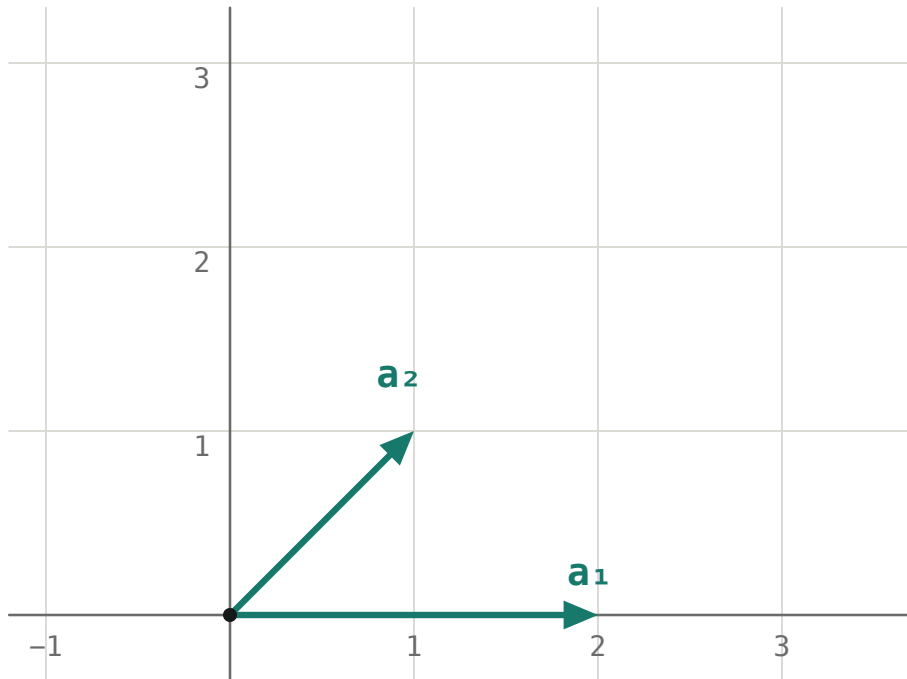
(B) Ax

(C) the candidate line

(D) every possible Ax

Least Squares

Perpendicular Columns: Which Gram Entries Change?



From Lecture 1: for $G = A^T A$ with columns a_1, a_2 , the entry is $G_{ij} = a_i^T a_j$.

a_2 is turned until perpendicular to a_1 and then doubled in length. What happens to G_{11}, G_{12}, G_{22} ?

- (A) unchanged, 0, $\times 4$
- (B) 1, 0, 1
- (C) unchanged, 0, $\times 2$
- (D) 0, unchanged, $\times 4$

DEFINITION

Least Squares

The Gram Matrix

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$$

$$G = A^T A \in \mathbb{R}^{n \times n}, \quad G_{ij} = a_i^T a_j$$

a_j column j of $A \in \mathbb{R}^{m \times n}$ — any A , any rank

G_{ij} row i , column j : column i dotted with column j

$G^T = G$ symmetric, because $a_i^T a_j = a_j^T a_i$

n the number of parameters, not of measurements

WORKED EXAMPLE

$$a_1 = (1, 1, 1), a_2 = (0, 1, 2)$$

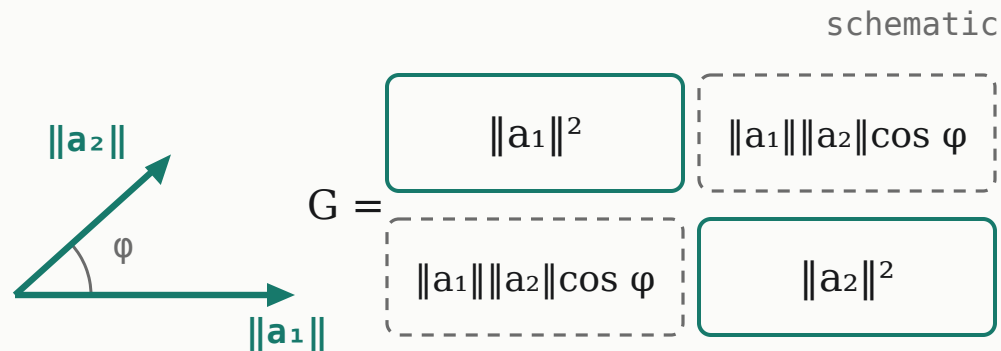
$G =$

$a_1^T a_1$ $1+1+1 = 3$	$a_1^T a_2$ $0+1+2 = 3$
$a_2^T a_1$ 3	$a_2^T a_2$ $0+1+4 = 5$

Boyd, S. and Vandenberghe, L. *Introduction to Applied Linear Algebra*, least squares. Cambridge, 2018.

Least Squares

Reading the Entries



$$G_{ii} = \|\mathbf{a}_i\|^2, \quad G_{ij} = \|\mathbf{a}_i\| \|\mathbf{a}_j\| \cos \varphi$$

G_{ii} squared length of column i

G_{ij} lengths **and** angle at once (p8)

φ between $\mathbf{a}_i, \mathbf{a}_j$, both $\neq 0$

Large $|G_{ij}|$ alone does not imply near-parallel columns. $G = I$ exactly for orthonormal columns, singular exactly for dependent ones (p55).

Break

10 minutes



The Gram matrix records every dot product between two columns of A .

Which equations does it give for the best-fit parameters?

For our three readings: $G = A^T A = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}$

Least Squares

The Normal Equations Are Necessary

NECESSARY

If $\hat{x}$ minimises, $r = b - A\hat{x} \perp \text{Col}(A)$ (p29)

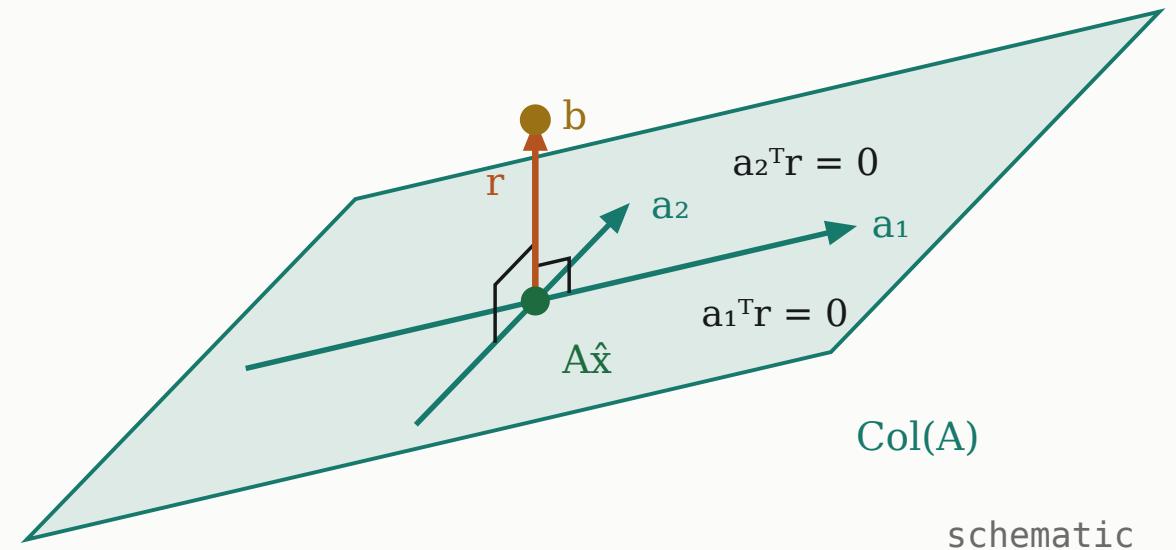
$\perp \text{Col}(A) \Leftrightarrow a_i^T r = 0$ for every column
(p14)

so $A^T r = 0: r \in \text{Null}(A^T)$ (p15)

$$A^T(b - A\hat{x}) = 0 \Rightarrow \boxed{A^T A\hat{x} = A^T b}$$

$A^T A$ $n \times n$, any A ; $A^T b \in \mathbb{R}^n$

necessary every minimiser solves it; the converse:
p50-51



Least Squares

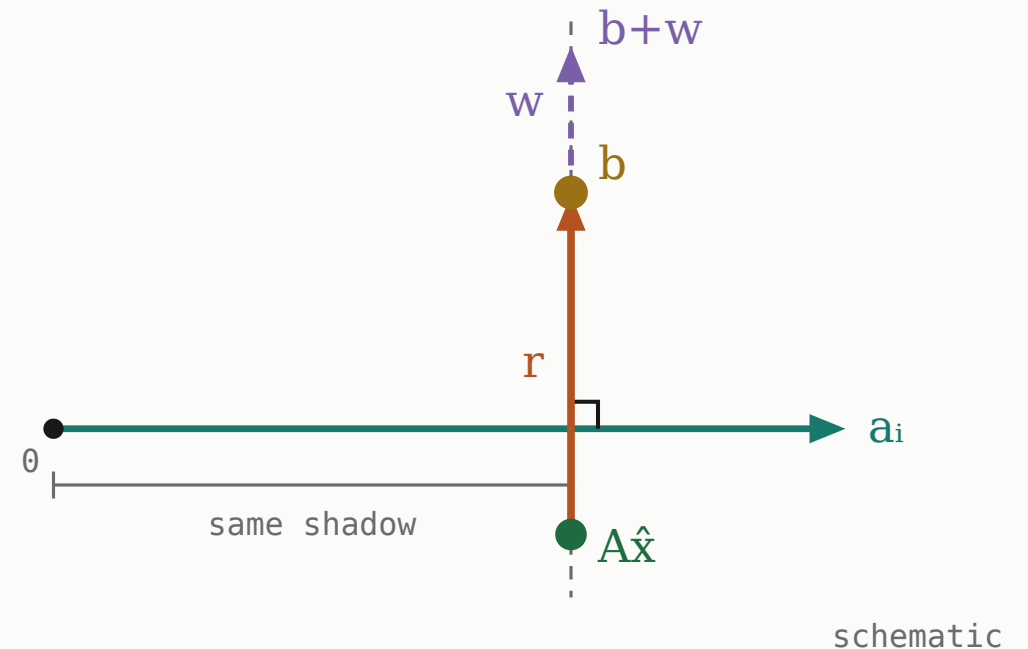
Reading the n Equations

$$a_i^\top (A\hat{x}) = \sum_j (a_i^\top a_j) \hat{x}_j = a_i^\top b$$

row i what column i measures is matched; the leftover is invisible to it

$n \times n$ n equations, n unknowns, any A ; solvable: take $Ax = \hat{b}$;
 $A^\top(b - \hat{b}) = 0$ (p29); only one? p54

Move b by any $w \in \text{Null}(A^\top) = \text{Col}(A)^\perp$ (p15): $A^\top b$, and the equations, do not change (p32).



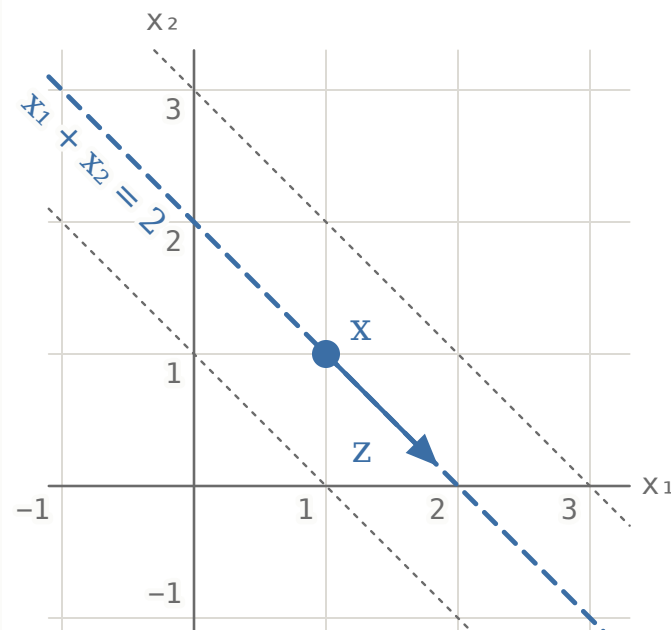
Least Squares

What Changes Along the Valley?

A is 3×2 , all ones; $b = (1, 2, 3)$. Move x along $z = (1, -1)$. Which change?

- ☐ (A) parameters x
- ☐ (B) prediction Ax
- ☐ (C) residual $r = b - Ax$
- ☐ (D) objective $f = \|Ax - b\|^2$

parameter space $\mathbb{R}^2$



output space $\mathbb{R}^3$

plane of $\text{Col}(A)$ and b



PREDICT

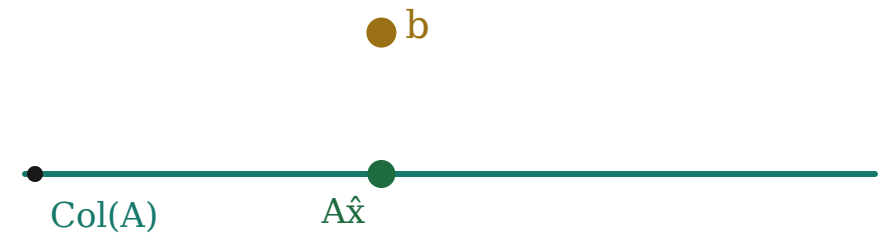
Least Squares

Are the Normal Equations Enough?

$\hat{x}$ solves $A^T A \hat{x} = A^T b$. For every A , which is guaranteed? $\hat{x}$ is...

- (A) a global minimiser
- (B) flat here, but lower somewhere else
- (C) the only minimiser

p49's A and b · output space $\mathbb{R}^3$



Least Squares

Why They Are Enough

SUFFICIENT

Take any other $\hat{x} + z$:

$$A(\hat{x} + z) - b = Az - r$$

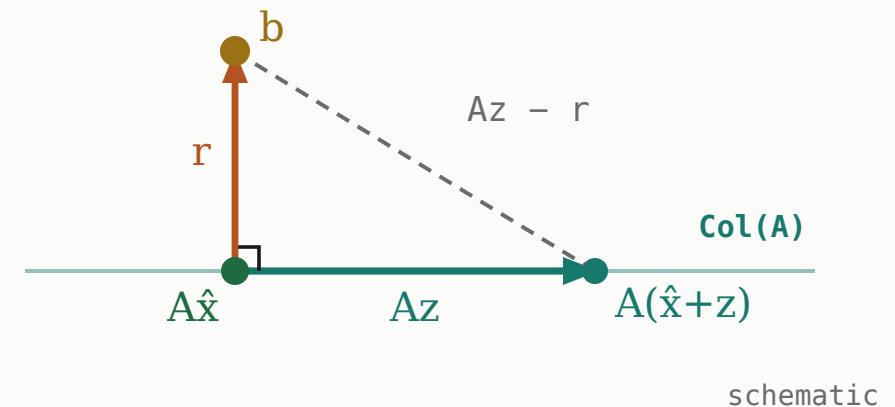
$z \in \mathbb{R}^n$ any step in parameter space; $r = b - A\hat{x}$

$$\|Az - r\|^2 = \|Az\|^2 - 2z^T(A^T r) + \|r\|^2$$

cross term $A^T r = 0$ is the equation $\hat{x}$ solves, so it is 0

$$\|A(\hat{x} + z) - b\|^2 = \|r\|^2 + \|Az\|^2 \geq \|r\|^2$$

both squares global minimum; equality exactly when $Az = 0$



Least Squares

The Whole Set of Minimisers

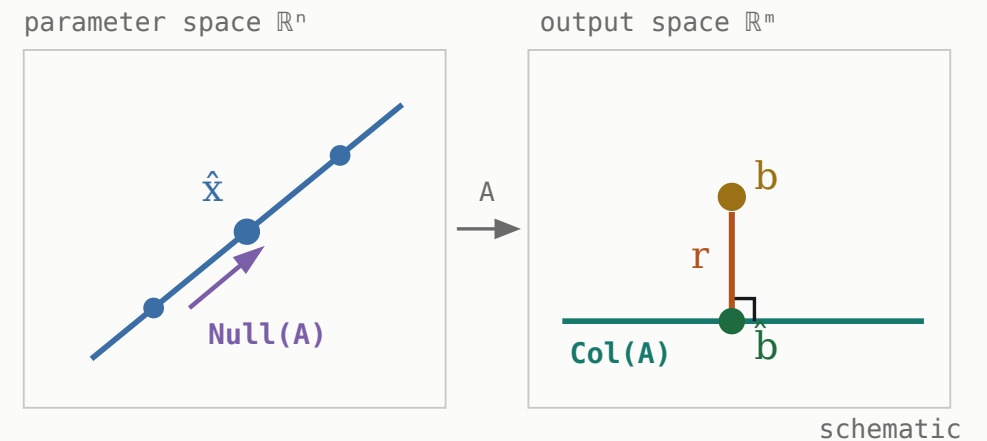
$$\hat{x} \text{ minimises } \iff A^T A \hat{x} = A^T b$$

every $A \Rightarrow$ p47 (necessary), $\Leftarrow$ p51 (enough); no rank assumption

$$\operatorname{argmin} f = \hat{x} + \operatorname{Null}(A)$$

Lecture 1 $x_p + \mathcal{N}(A)$; all share one $\hat{b}$, one r

p49: $\hat{x} = (1, 1)$, $z = (1, 0) \notin \operatorname{Null}(A)$: $Az = (1, 1, 1)$, $f = 2 + 3 = 5$.

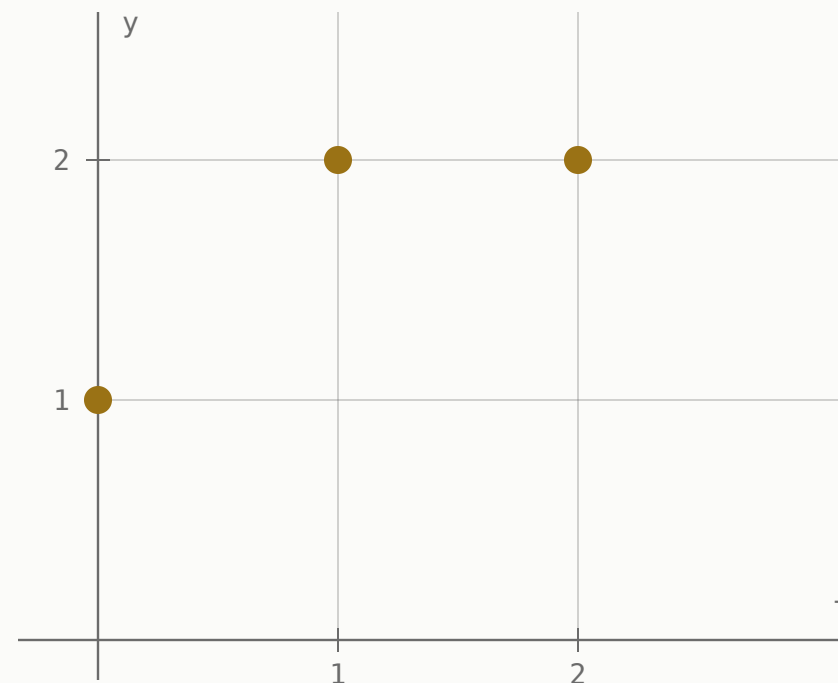


$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

Least Squares

Solve the Three-Point Fit

$G = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}$, $A^T b = (5, 6)$. Predict the signs of r , then find $\hat{x}, \hat{b}, r$.



Least Squares

When Is the Minimiser Unique?

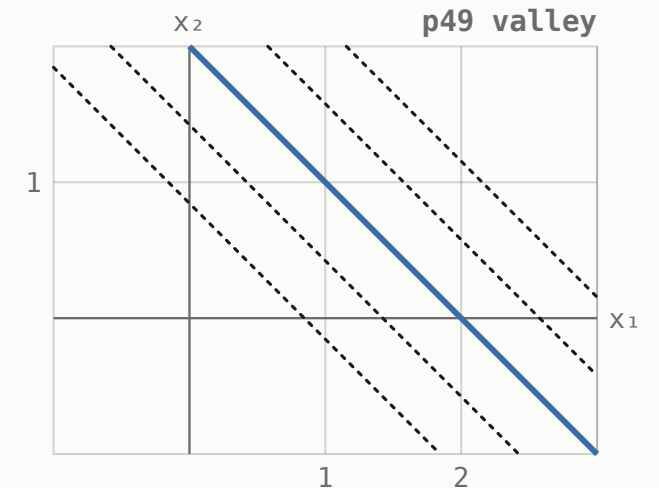
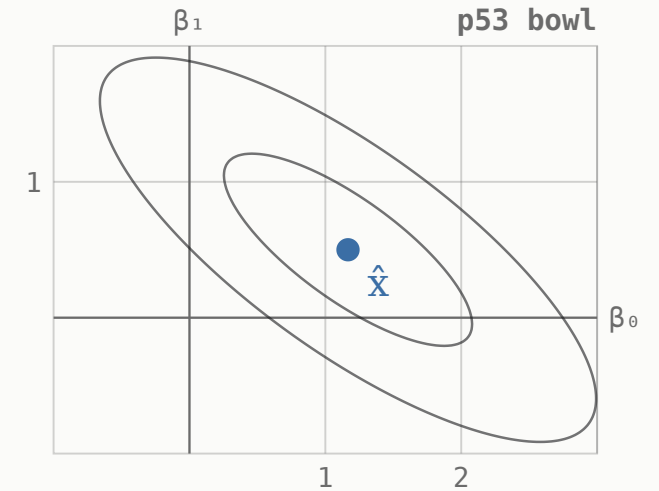
$$\hat{x} + \text{Null}(A) = \{\hat{x}\} \iff \text{Null}(A) = \{0\} \iff \text{rank } A = n$$

argmin $= \hat{x} + \text{Null}(A)$ (p52); one point $\Leftrightarrow$ independent columns

rank-nullity $\dim \text{Null}(A) = n - \text{rank } A$

$m \geq n$ from $\text{rank } A = n \leq m$; not assumed

p53's fit: independent columns, one bottom. p49: a valley of bottoms, one $\hat{b}$.



Least Squares

Gram Positivity and the Equivalent Conditions

$$z^T G z = z^T A^T A z = \|A z\|^2 \geq 0$$

$$G = A^T A \text{ symmetric } n \times n, \text{ any } A$$

Positive definite: symmetric G , $z^T G z > 0$ for $z \neq 0$.

$$\text{Null}(A) = \{0\} \iff G \text{ pos. definite} \text{ — } \|A z\|^2 > 0$$

$$\iff G \text{ invertible} \text{ — } A z = 0 \Rightarrow G z = 0, G z = 0 \Rightarrow A z = 0: \text{Null}(G) = \text{Null}(A)$$

$$\iff \text{one } \hat{x} \text{ — argmin} = \hat{x} + \text{Null}(A) \text{ (p52)}$$

Fails: G singular, flat valley.

Boyd and Vandenberghe. *Introduction to Applied Linear Algebra*, least squares. Cambridge, 2018.

Least Squares

Closed Forms Under Full Column Rank

RANK $A = n$ (P54)

$$\hat{x} = (A^T A)^{-1} A^T b, \quad P = A(A^T A)^{-1} A^T$$

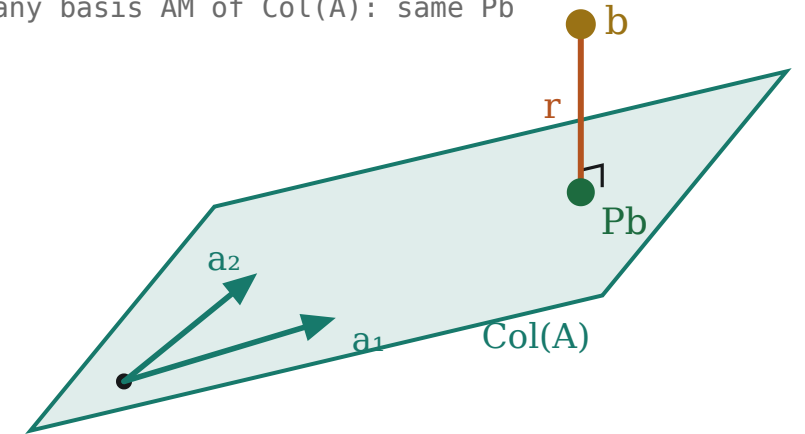
$\hat{x} \in \mathbb{R}^n$ $A^T A$ invertible (p55); in code, solve the system

P $m \times m$, symmetric, rank n ; rank $< n$: next lecture

$Pb \in \text{Col}(A)$, $A^T(b - Pb) = 0$: **the** projection (p29) — P depends only on $\text{Col}(A)$.

$A = Q$ orthonormal: $Q^T Q = I$, $P = Q Q^T$ (p30). Otherwise the Gram factor corrects the basis.

any basis AM of $\text{Col}(A)$: same Pb



schematic

Gram-Schmidt Orthogonalization

Why We Want an Orthonormal Basis

ANY BASIS

$$P = A(A^T A)^{-1} A^T$$

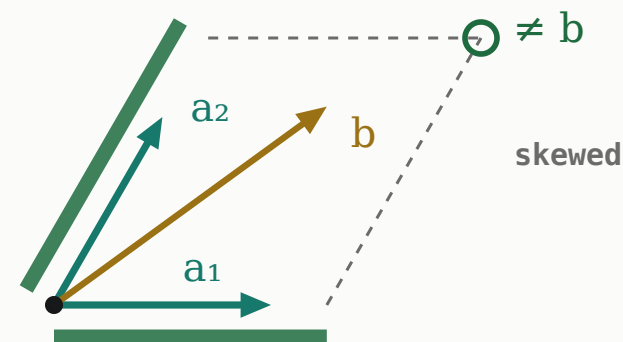
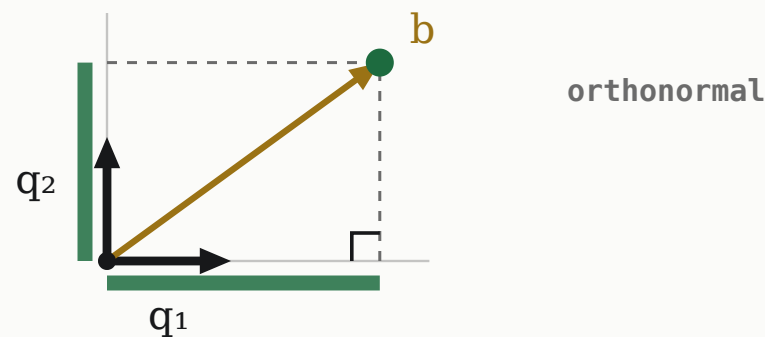
cost build $A^T A$, then solve

ORTHONORMAL BASIS

$$P = Q Q^T$$

$Q^T b$ coordinates of $Pb = \sum_i q_i (q_i^T b)$: one each

Can we build such a Q for the same $\text{Col}(A)$?



Gram-Schmidt Orthogonalization

Which Subtraction Makes It Orthogonal?

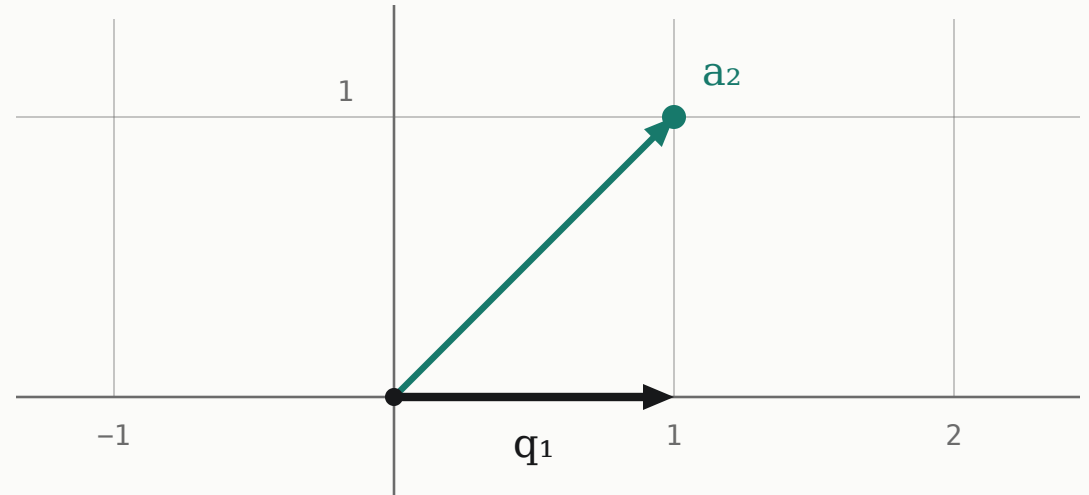
$q_1 = (1, 0)$, $a_2 = (1, 1)$. Sketch each result: which removes exactly the part along q_1 ?

(A) $a_2 - q_1(q_1^T a_2)$

(B) $a_2 - q_1\|a_2\|$

(C) $a_2/\|a_2\|$

(D) $a_2 + q_1(q_1^T a_2)$



$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Gram-Schmidt Orthogonalization

Gram-Schmidt on Two Vectors

THE COLUMNS FROM LECTURE 2

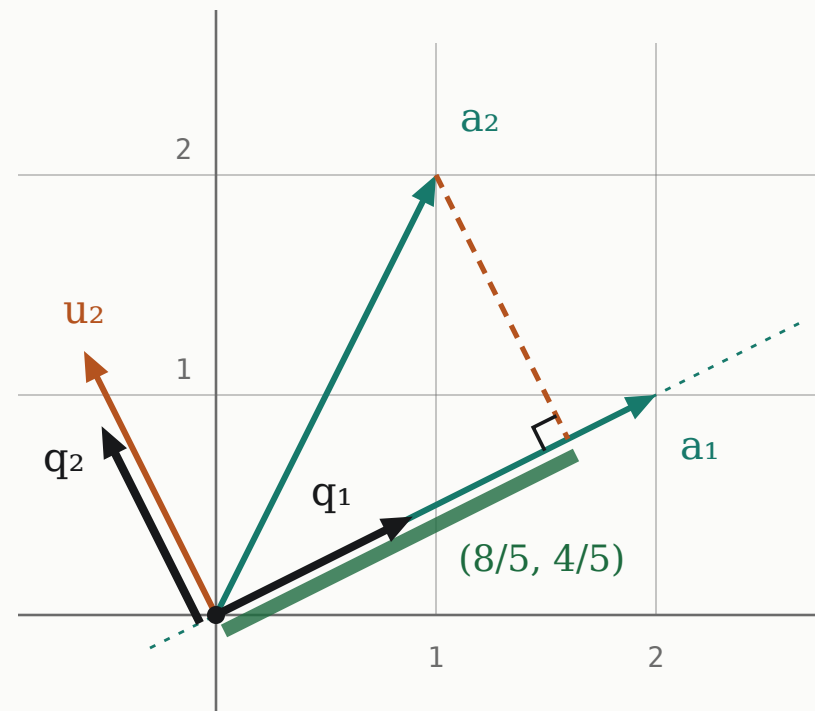
$$a_1 = (2, 1), a_2 = (1, 2)$$

NORMALISE $\|a_1\| = \sqrt{5} \neq 0, q_1 = (2, 1)/\sqrt{5}$

COMPONENT $q_1^T a_2 = 4/\sqrt{5}$ (becomes r_{12})

REMAINDER $u_2 = (1, 2) - (8/5, 4/5) = (-3/5, 6/5)$

NORMALISE $\|u_2\| = 3/\sqrt{5} \neq 0, q_2 = (-1, 2)/\sqrt{5}$



DEFINITION

Gram-Schmidt Orthogonalization

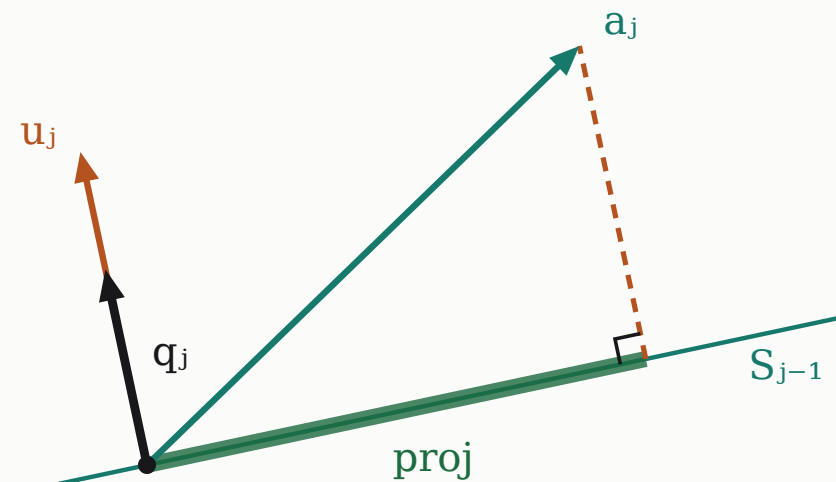
Keep the New Direction, Remove the Overlap

$$u_j = a_j - \sum_{i < j} q_i (q_i^\top a_j), \quad q_j = \frac{u_j}{\|u_j\|}$$

S_{j-1} span of the orthonormal $q_1, \dots, q_{j-1}$; $j = 1$: empty sum, $S_0 = \{0\}$

$\sum$ = projection onto S_{j-1} only because the q_i are orthonormal

$u_j \in \mathbb{R}^m$ remainder; dividing needs $u_j \neq 0$ ($j = 1$: $u_1 = a_1 \neq 0$)



Axler. *Linear Algebra Done Right*, 4th ed., 6B. Springer, 2024.

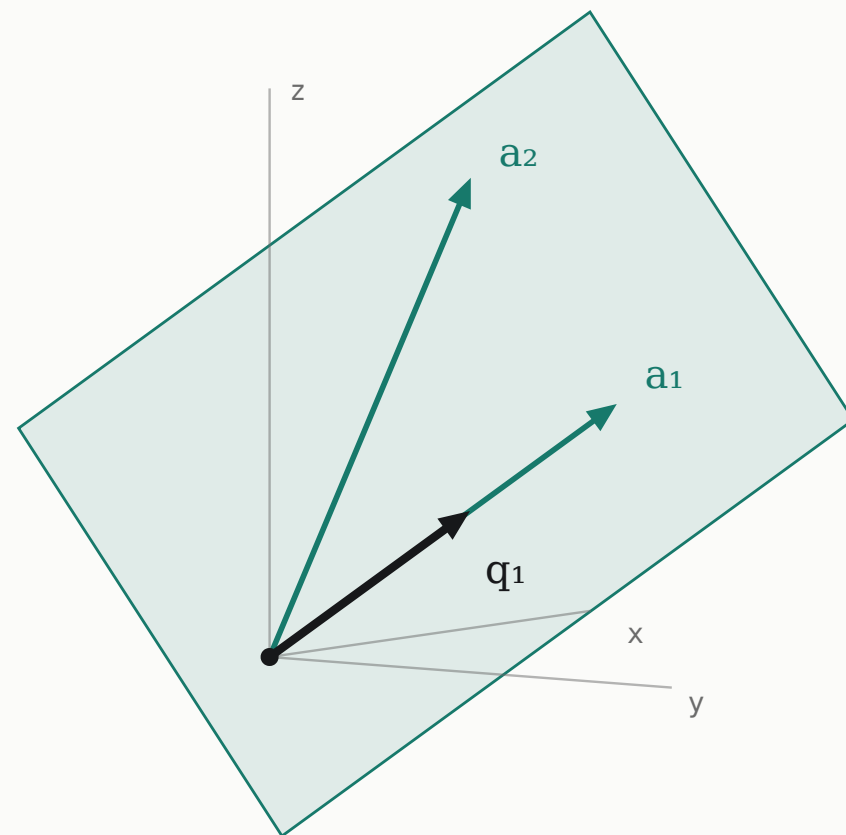
$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$$

Gram-Schmidt Orthogonalization

Orthogonalize Our Fitting Matrix

$q_1 = (1, 1, 1)/\sqrt{3}$, $u_2 \in \text{span}\{a_1, a_2\}$.

Predict its entries and the checks.



Gram-Schmidt Orthogonalization

The General Procedure

Scope: $a_1, \dots, a_n$ independent — A of full column rank.

for $j = 1, \dots, n$:

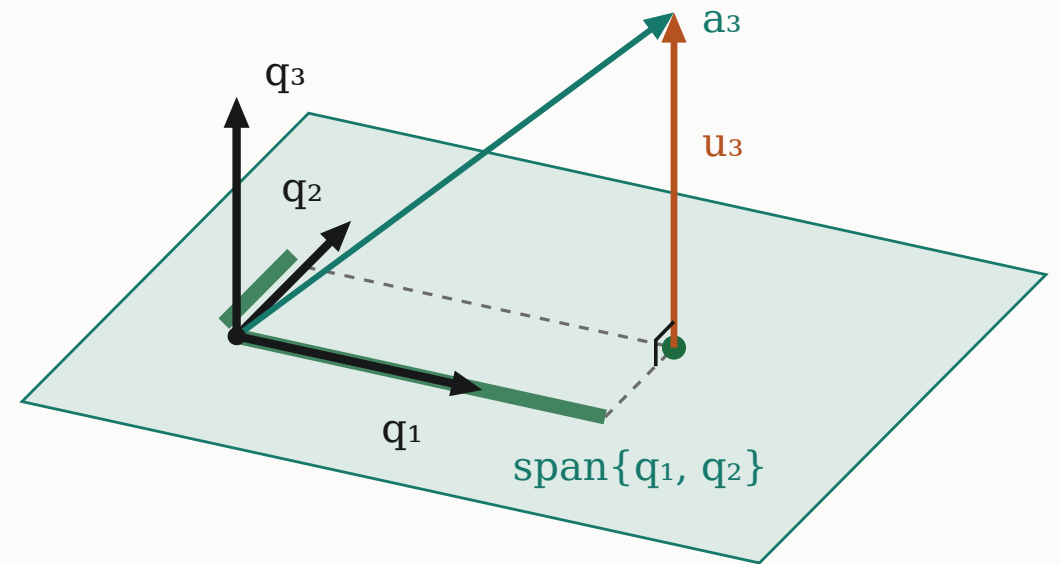
$$u_j = a_j - \sum_{i < j} q_i (q_i^\top a_j)$$

$$q_j = u_j / \|u_j\|$$

a_j column j of $A \in \mathbb{R}^{m \times n}$

$q_i^\top a_j$ coefficient on earlier q_i

Under that scope $\|u_j\| > 0$ at every step, so each division is legal.



Gram-Schmidt Orthogonalization

Why the Result Is Orthogonal

$$q_i^\top u_j = q_i^\top a_j - \sum_{l < j} (q_i^\top q_l)(q_l^\top a_j)$$

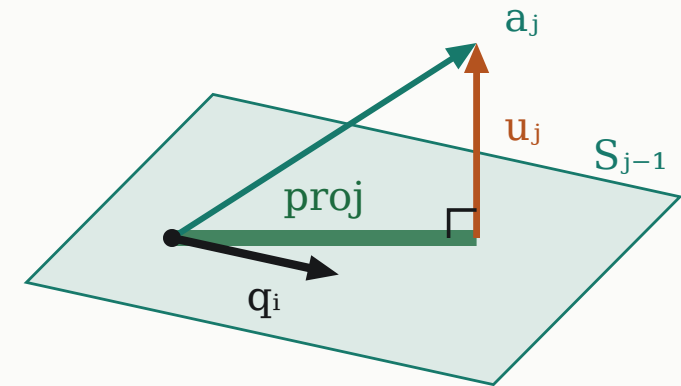
assume $q_1, \dots, q_{j-1}$ already orthonormal — the induction

$i < j$ $q_i^\top q_l = 1$ at $l = i$, else 0: one term survives

u_j the remainder before normalising, in $\mathbb{R}^m$

$$= q_i^\top a_j - q_i^\top a_j = 0 \text{ for every } i < j.$$

$q_j = u_j / \|u_j\|$, so $q_i^\top q_j = 0$ too: scaling cannot change a zero.



Gram-Schmidt Orthogonalization

Why the Span Is Unchanged

After subtracting the projection, can we still rebuild a_j ?

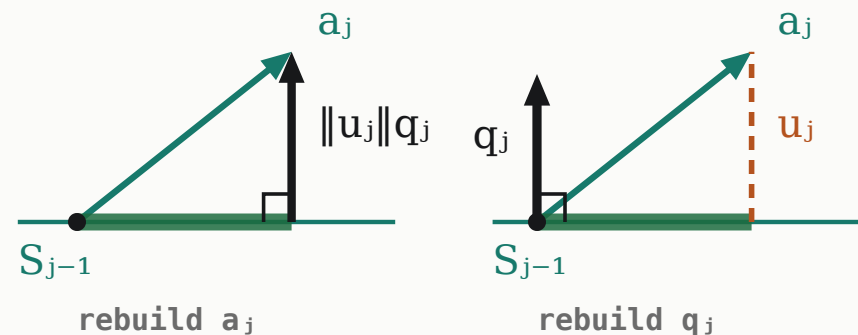
$$a_j = \sum_{i < j} (q_i^T a_j) q_i + \|u_j\| q_j$$

$\|u_j\| > 0$ by the scope

$\text{span}\{q_1..q_j\} = \text{span}\{a_1..a_j\}$

$\supseteq$: this identity at each step $i \leq j$

$\subseteq$: solve for q_j : from a_j and earlier q 's (induction)



PREDICT

Gram-Schmidt Orthogonalization

What If Nothing Remains?

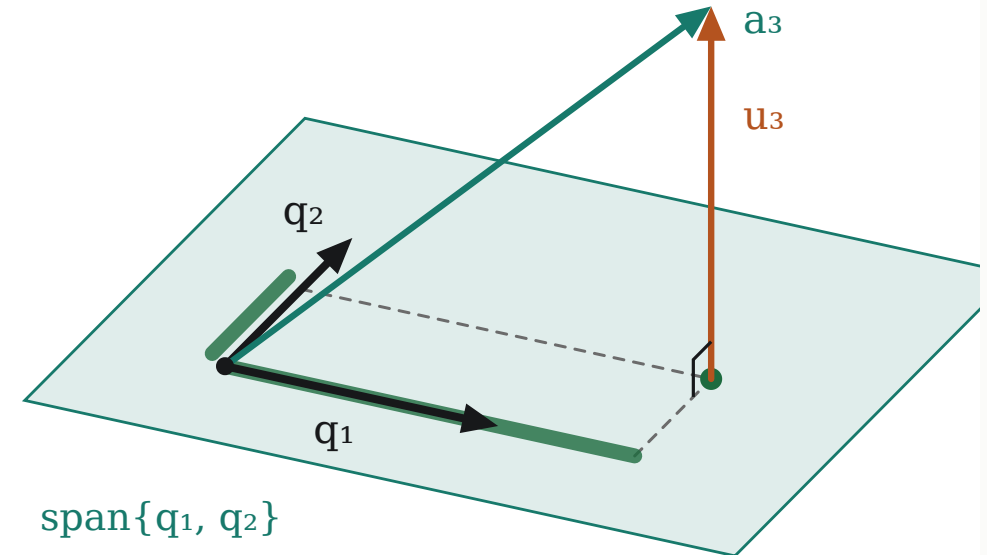
Lower a_3 into the plane: which hits 0, a_3 or u_3 ?

$u_3 = 0$ means:

- (A) $a_3 \perp q_1, q_2$
- (B) $a_3 \in \text{span}\{q_1, q_2\}$
- (C) $a_3 = 0$
- (D) q_1, q_2 invalid

$$\|a_3\| = 2.14$$

$$\|u_3\| = 1.40$$



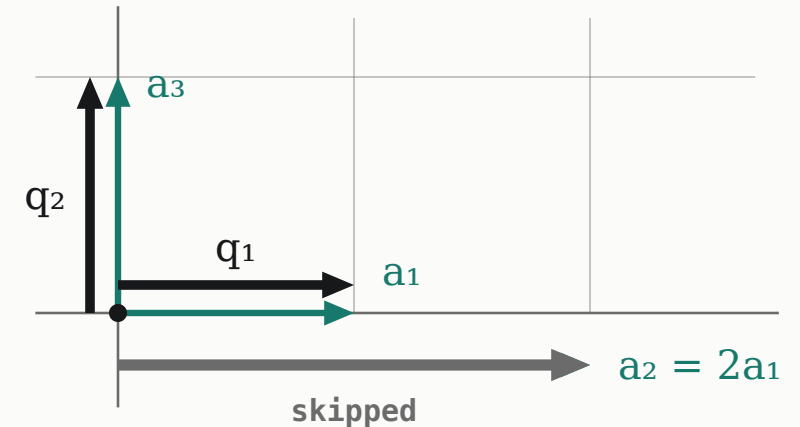
Gram-Schmidt Orthogonalization

What the Procedure Does Not Do

Why we could divide: $u_j = 0$ would write a_j from the earlier q 's, which independence forbids.

Without it, halting at $u_j = 0$ is **not** basis extraction: $[e_1, 2e_1, e_2]$ stops at $j = 2$, missing e_2 .

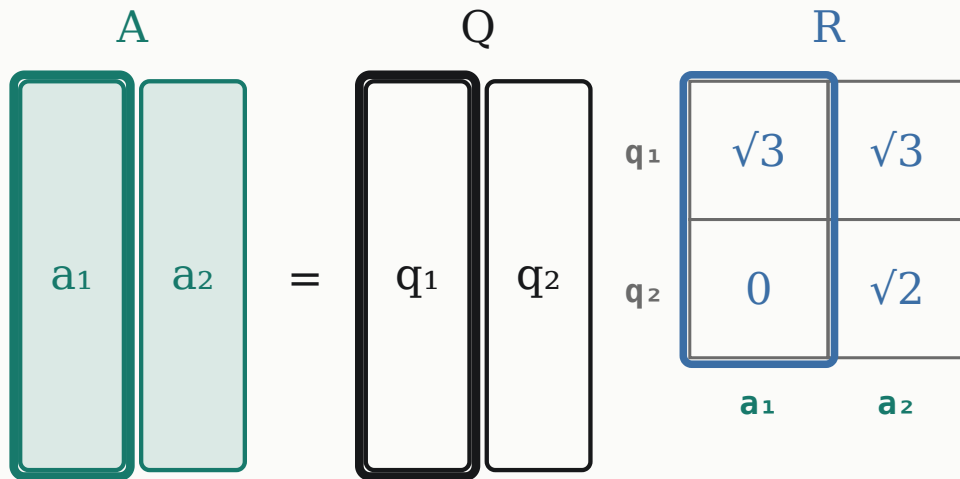
Skip it, carry on in the original order, index the kept q 's.
Pivoting instead reorders columns: $A\Pi = QR$ (Lecture 4).



QR Factorization and Least Squares

Record the Coefficients: $A = QR$

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \quad Q = \begin{bmatrix} 1/\sqrt{3} & -1/\sqrt{2} \\ 1/\sqrt{3} & 0 \\ 1/\sqrt{3} & 1/\sqrt{2} \end{bmatrix}$$



$$a_j = \sum_{i \leq j} r_{ij} q_i$$

$$r_{ij} = q_i^\top a_j, \quad i < j$$

$$r_{jj} = q_j^\top a_j = \|u_j\| > 0: \text{columns independent}$$

$$Q^\top A = (Q^\top Q)R = R$$

$$Q^\top Q = I_n \text{ (p19)}$$

R column j : a_j in the q -basis

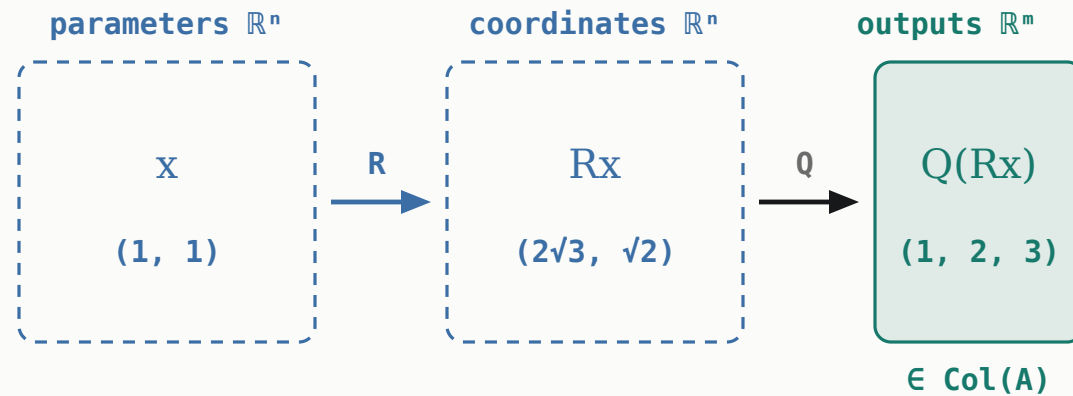
WORKED EXAMPLE

$$q_1^\top a_2 = \sqrt{3}, \quad \|u_2\| = \sqrt{2}; \text{ check } \sqrt{3}q_1 + \sqrt{2}q_2 = a_2$$

$$R = \begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{2} \end{bmatrix} \quad G = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}$$

QR Factorization and Least Squares

What $Ax = Q(Rx)$ Means



$$Ax = Q(Rx)$$

x parameters, $\in \mathbb{R}^n$

Rx Ax in the q -basis, also $\in \mathbb{R}^n$

Q keeps lengths (p19); R need not

$$R^T R = A^T A = G$$

always $R^T R = (QR)^T QR$

here $G \neq I$: R is not a rotation

QR Factorization and Least Squares

Why Is R Upper Triangular?

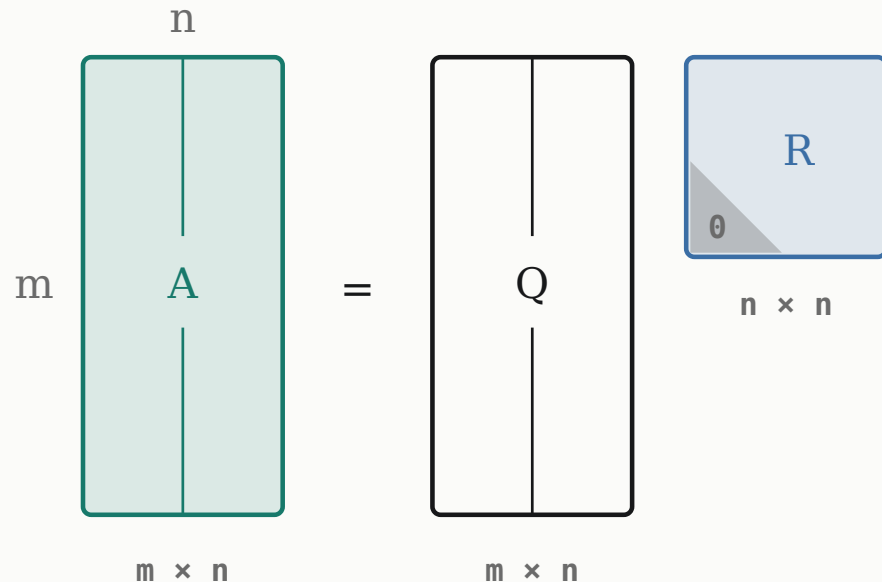
	a_1	a_2	a_3
q_1			
q_2			
q_3			

a_j uses only $q_1, \dots, q_j$ (p67). Why is $r_{ij} = q_i^T a_j = 0$ for $i > j$?

- (A) The columns of A are orthogonal
- (B) $q_i \perp q_1, \dots, q_j$, so $\perp$ their span
- (C) Orthonormal Q forces triangular R
- (D) Only when A is square

QR Factorization and Least Squares

Read the Shapes Before the Entries



Here $m \geq n$ and A has full column rank.

$$Q \in \mathbb{R}^{m \times n}, R \in \mathbb{R}^{n \times n}, Q^T Q = I_n$$

thin Q is as tall as A (p19)

shapes $(m \times n)(n \times n) = (m \times n)$

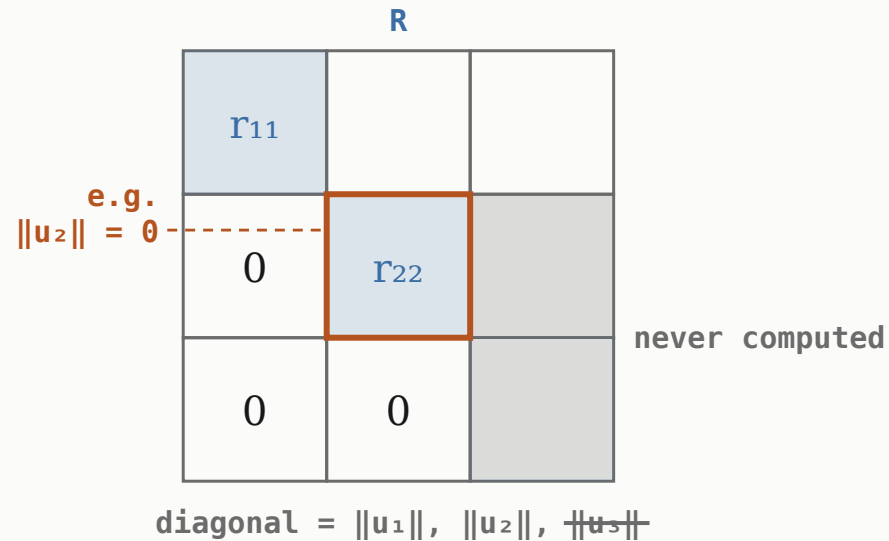
$QQ^T \neq I_m$ once $n < m$ (p18)

We assume full column rank so this construction cannot break down and R is invertible.

$$R = \begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{2} \end{bmatrix}$$

QR Factorization and Least Squares

Why R Is Invertible



$$\det R = r_{11} \cdots r_{nn} \neq 0$$

$r_{jj} > 0$ $r_{jj} = \|u_j\|$, and $u_j = 0$ would put a_j in the span of the earlier columns (p66)

$\neq 0$ triangular; ours $\sqrt{3}\sqrt{2} = \sqrt{6}$

Rank $< n$: Gram-Schmidt stalls; any R has an $r_{jj} = 0$, so $\det R = 0$ and the solve divides by zero.

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

QR Factorization and Least Squares

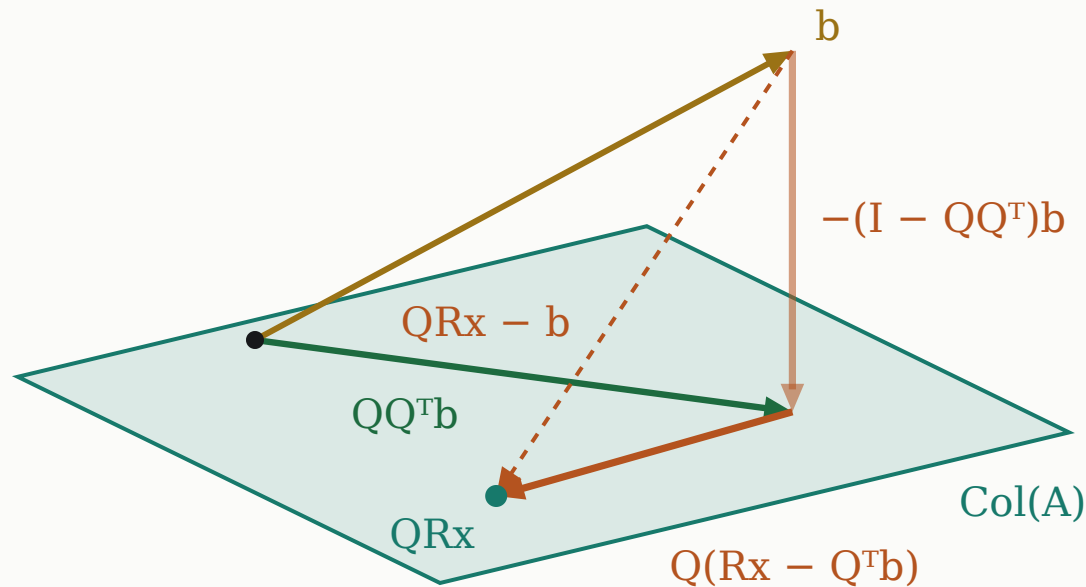
QR Is Not Diagonalization

	$A = V\Lambda V^{-1}$	$A = QR$
DIRECTIONS	eigenvectors	orthonormal columns
MIDDLE	diagonal	upper triangular
USE	modes, powers	outputs, least squares
NEEDS SQUARE A	yes	no

Is q_1 an eigenvector of A ?

QR Factorization and Least Squares

Split the Error in the New Coordinates



Throughout $r = b - Ax$, so $QRx - b = -r$ and $\|QRx - b\| = \|r\|$: the same error, drawn from b to the prediction.

Do not start from $QRx = b$: generally false.

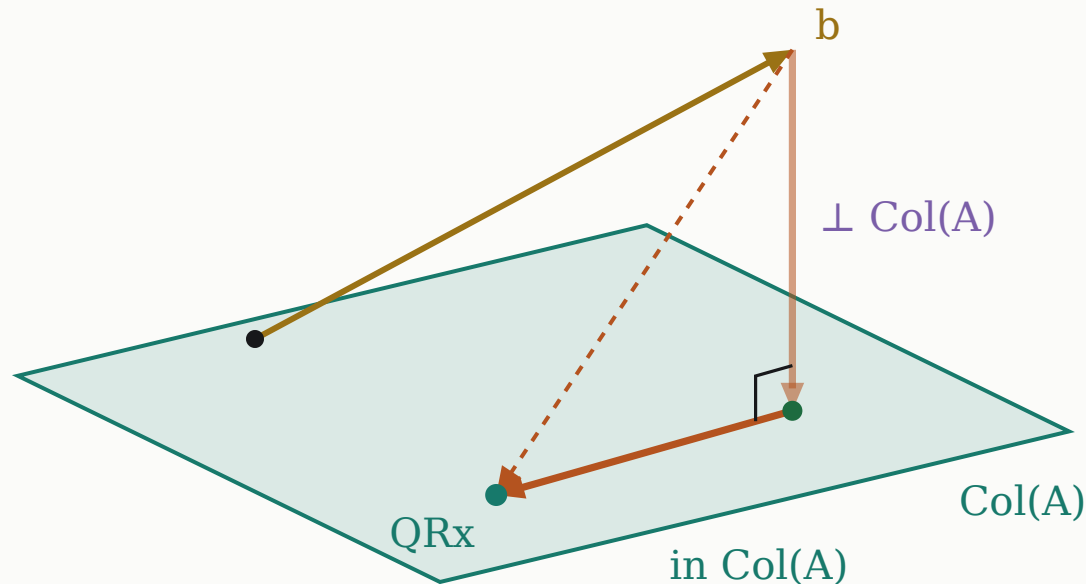
$$QRx - b = Q(Rx - Q^T b) - (I - QQ^T)b$$

$$\text{check} = QRx - QQ^T b - b + QQ^T b$$

$$\text{goal} \min_x \|r\|^2$$

QR Factorization and Least Squares

Why the Two Parts Are Perpendicular



schematic

The two pieces of $QRx - b$:

$$Q(Rx - Q^T b) \in \text{Col}(Q) = \text{Col}(A)$$

= the span was kept

$$Q^T(I - QQ^T)b = Q^T b - Q^T b = 0$$

$\Rightarrow \perp$ every q_i , so $\perp \text{Col}(A)$

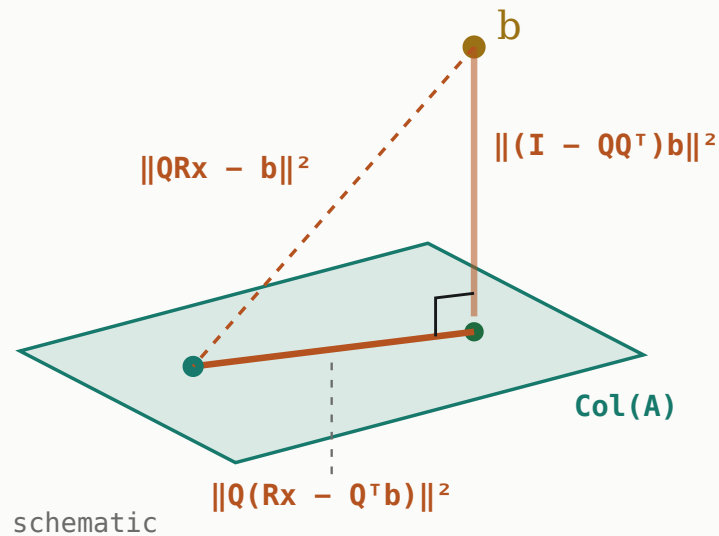
$(I - QQ^T)b = (I - P)b$, the projection's residual; drawn
leg $-(I - P)b$

The same perpendicular-error decomposition, in the same colours.

QR Factorization and Least Squares

Pythagoras in the New Coordinates

outputs $\mathbb{R}^n$



coordinates $\mathbb{R}^n$

$$z = Rx - Q^T b$$

$$\|Qz\| = \|z\|$$

The two parts are perpendicular (p74):

$$\|QRx - b\|^2 = \|Q(Rx - Q^T b)\|^2 + \|(I - QQ^T)b\|^2$$

+ cross term = 0: squares add (p11)

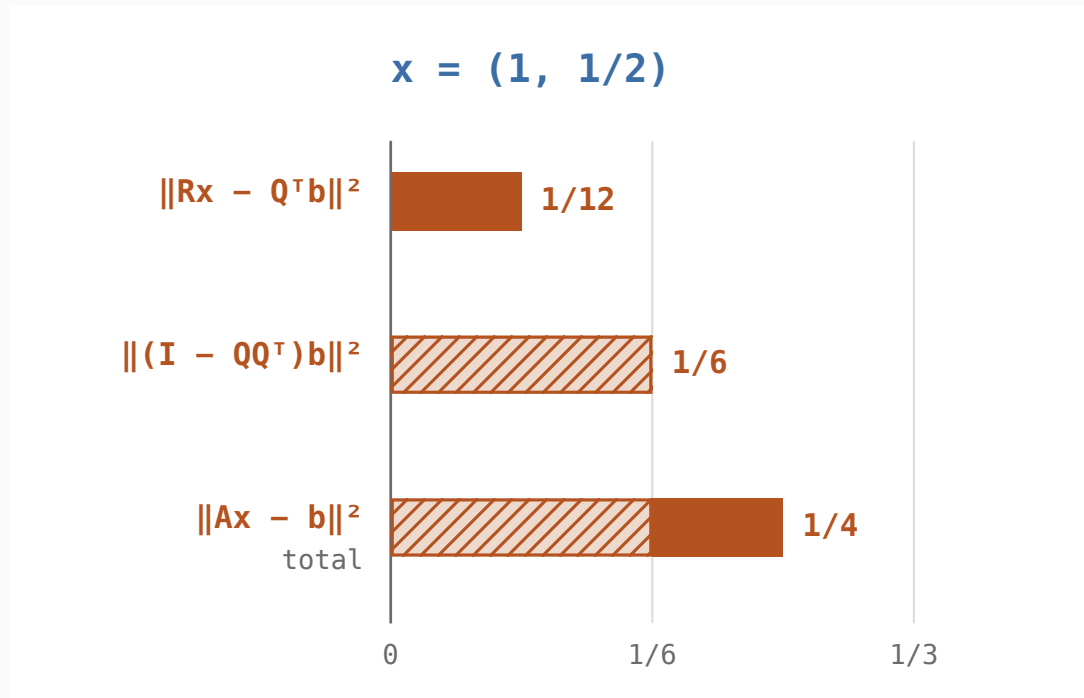
$$\|Ax - b\|^2 = \|Rx - Q^T b\|^2 + \|(I - QQ^T)b\|^2$$

$$z = Rx - Q^T b \quad \|Qz\|^2 = z^T Q^T Q z = \|z\|^2 \text{ (p19)}$$

QR Factorization and Least Squares

Adjustable and Unavoidable Error

$$b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$



At the solution, which part must be zero?

$$\|Ax - b\|^2 = \|Rx - Q^T b\|^2 + \|(I - QQ^T)b\|^2$$

QR Factorization and Least Squares

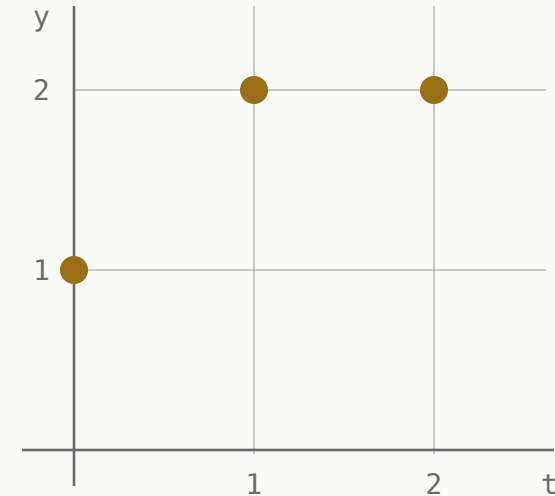
Solve the Same Fit with QR

Which unknown comes first, and why?

$$\begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{2} \end{bmatrix} \hat{x} = \begin{bmatrix} 5/\sqrt{3} \\ 1/\sqrt{2} \end{bmatrix}$$

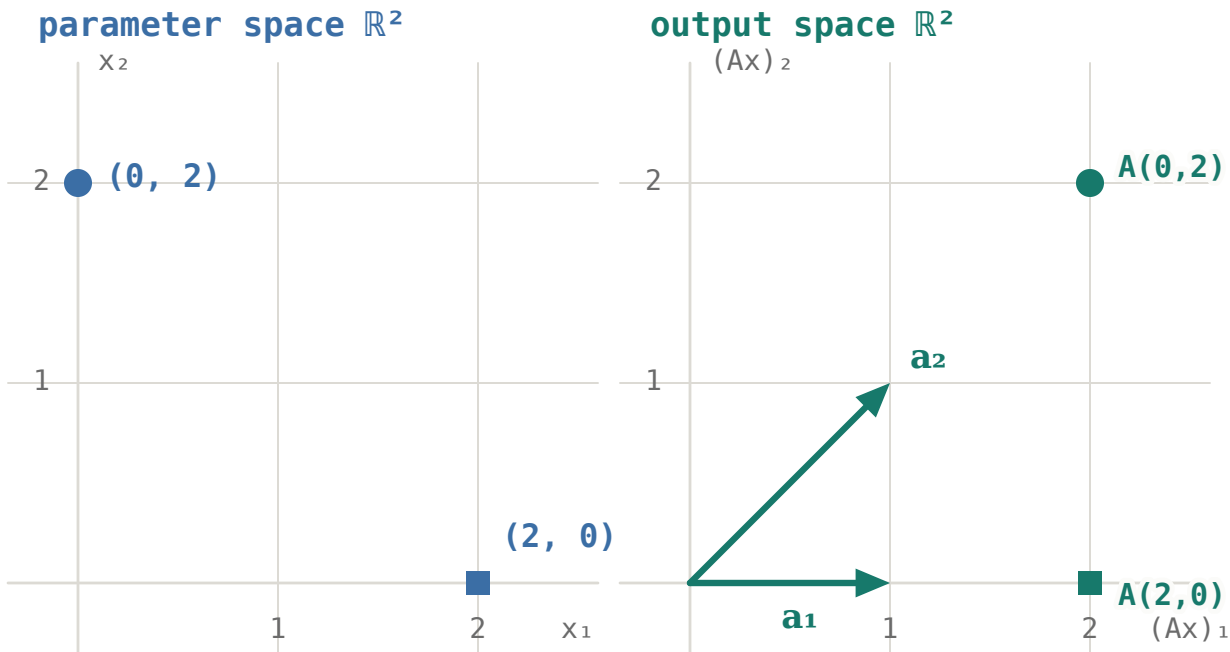
$\hat{x}$ (β_0, β_1)

R $r_{jj} > 0$



QR Factorization and Least Squares

Very Different Parameters, Almost the Same Prediction



What becomes hard when we infer parameters from measurements?

QR Factorization and Least Squares

Largest Stretch over Smallest Stretch

$$\kappa_2(A) = \frac{\max_{\|z\|=1} \|Az\|_2}{\min_{\|z\|=1} \|Az\|_2}$$

A $m \times n$, $m \geq n$

z parameter direction, $\|z\|_2 = 1$

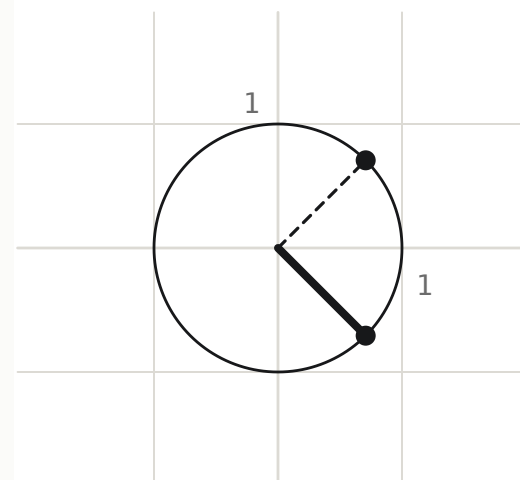
The second lecture measured how a matrix stretches directions; the new object is the ratio.

$\min > 0 \Leftrightarrow \text{Null}(A) = \{0\}$: no unit z is flattened.

Rank lost: some unit z has $Az = 0$, so $\min = 0$ and $\kappa_2(A) = \infty$ by convention.

$a_1 = (1, 0)$, $a_2 = (\cos \varphi, \sin \varphi)$, $c = \cos \varphi$
 $z = (\cos \psi, \sin \psi)$, $\varphi = 30.0^\circ$

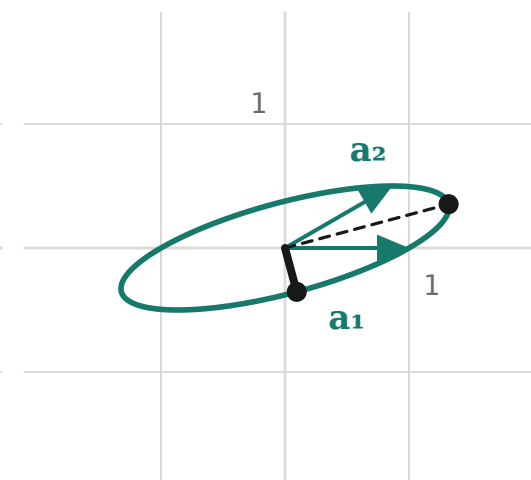
parameters: unit z



$\|z\| = 1$

dashed: most stretched z
thick: least stretched z

outputs: Az



$\|Az\|^2 = 1 + c \sin 2\psi$
 $\in [1 - c, 1 + c]$

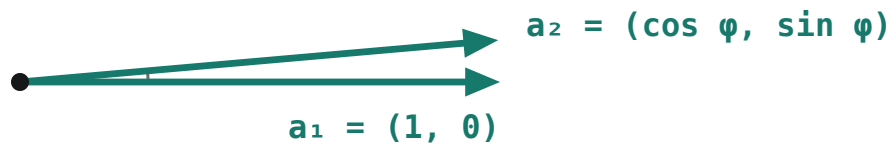
dashed $\sqrt{1 + c}$ · thick $\sqrt{1 - c}$
 $\kappa_2(A) = \sqrt{(1+c)/(1-c)} = 3.7$

DISCUSS

QR Factorization and Least Squares

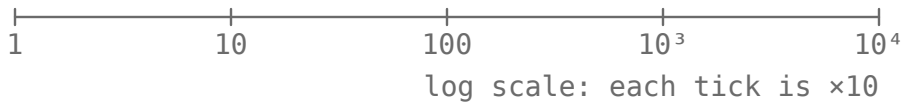
Which Computational Route Would You Choose?

$$\varphi = 5.0^\circ$$



$\kappa_2(A)$

22.9



Which is correct?

- ☐ (A) QR cures the sensitivity
- ☐ (B) Forming $A^T A$ can hurt the numerics
- ☐ (C) Normal equations fail even exactly
- ☐ (D) Every QR code is equally stable

QR Factorization and Least Squares

Squaring the Ratio

$$\kappa_2(R) = \kappa_2(A), \quad \kappa_2(A^T A) = \kappa_2(A)^2$$

R $\|Rz\| = \|QRz\| = \|Az\|$, Q an isometry

$A^T A$ full column rank; proved next lecture

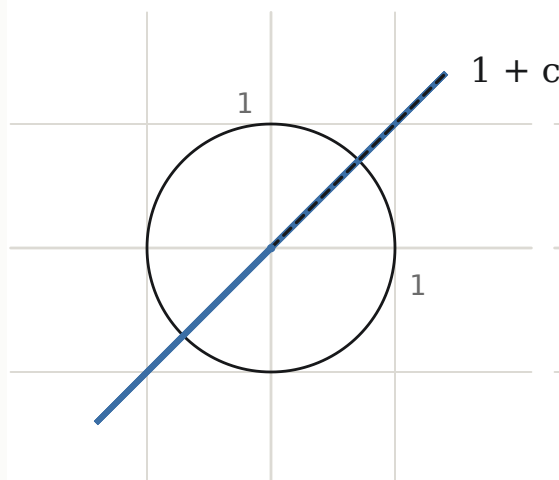
The triangular coefficient matrix has A 's stretch ratio; the normal equations square it.

Rules of thumb, not bounds: about $\log_{10} \kappa_2$ digits lost; sensitivity also depends on the residual, so $\kappa_2(A)$ alone does not settle it.

$$A = \begin{bmatrix} 1 & c \\ 0 & \sin \varphi \end{bmatrix} \quad \begin{array}{l} \mathbf{a}_1 = (1, 0), \quad \mathbf{a}_2 = (\cos \varphi, \sin \varphi), \quad c = \cos \varphi \\ z = (\cos \psi, \sin \psi), \quad \varphi = 5.0^\circ \end{array}$$

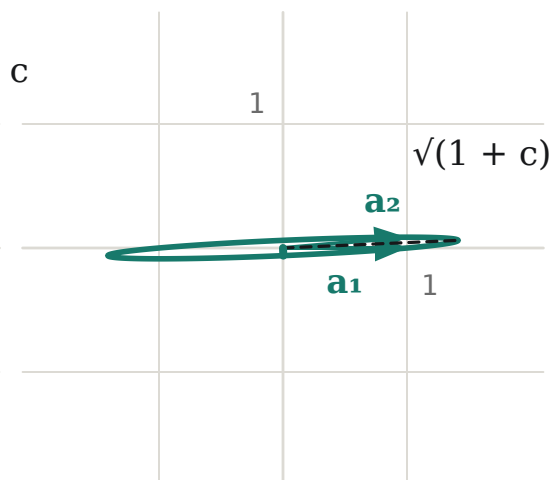
parameters: z , $A^T A z$

outputs: Az



$$\|A^T A z\|^2 = 1 + c^2 + 2c \sin 2\psi \\ \in [(1 - c)^2, (1 + c)^2]$$

$$\kappa_2(A^T A) = (1+c)/(1-c) = 525$$



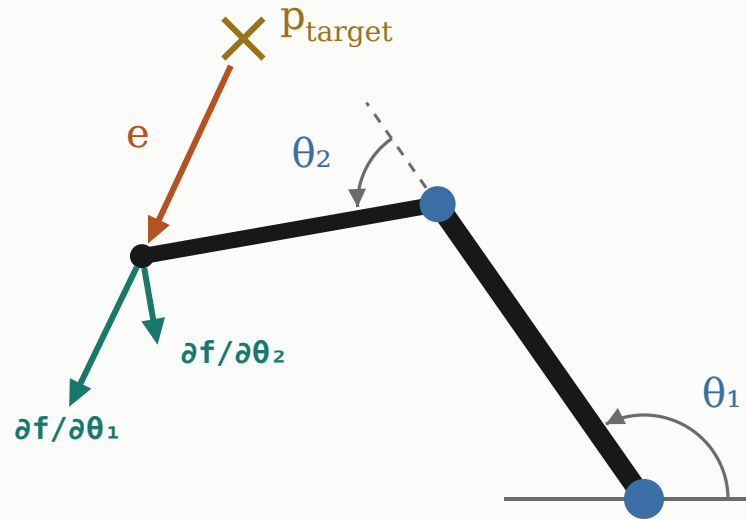
$$\|Az\|^2 = 1 + c \sin 2\psi \\ \in [1 - c, 1 + c]$$

$$\kappa_2(A) = \sqrt{((1+c)/(1-c))} = 22.9$$

Trefethen, L. N. and Bau, D. **Numerical Linear Algebra**, Lecture 18 conditioning of least squares problems. SIAM, 1997.

Robotics Application

The Local Model



$$e(\theta) = f(\theta) - p_{\text{target}} \in \mathbb{R}^2$$

$$e(\theta + \Delta\theta) \approx e(\theta) + J(\theta)\Delta\theta$$

f tip position, nonlinear in θ

$\theta = (\theta_1, \theta_2)$ joint angles in radians; $\Delta\theta$ small

$J(\theta)$ $\partial f / \partial \theta$; column j : tip velocity per unit joint rate

$J\Delta\theta$ a displacement, not a velocity

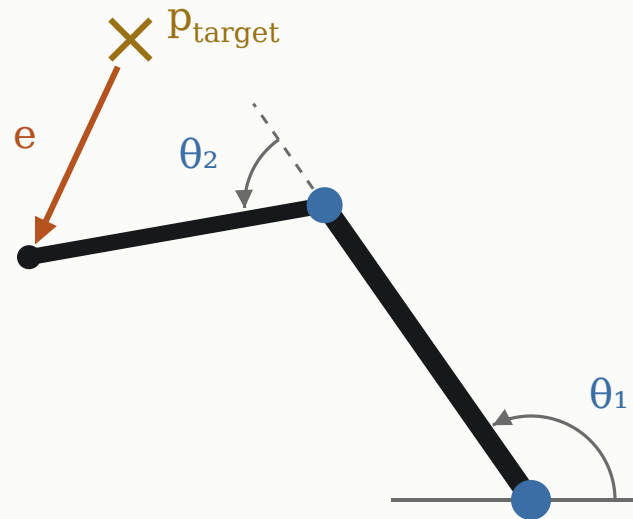
Near a singularity J can be ill-conditioned; at a singularity it loses rank.

$$\det J = l_1 l_2 \sin \theta_2, \text{ link lengths } l_1, l_2 > 0$$

Lynch, K. M. and Park, F. C. *Modern Robotics*, 5.3 singularities and 6.2 numerical inverse kinematics. Cambridge, 2017.

Robotics Application

One Least-Squares Step

 $A =$ $x =$ $b =$

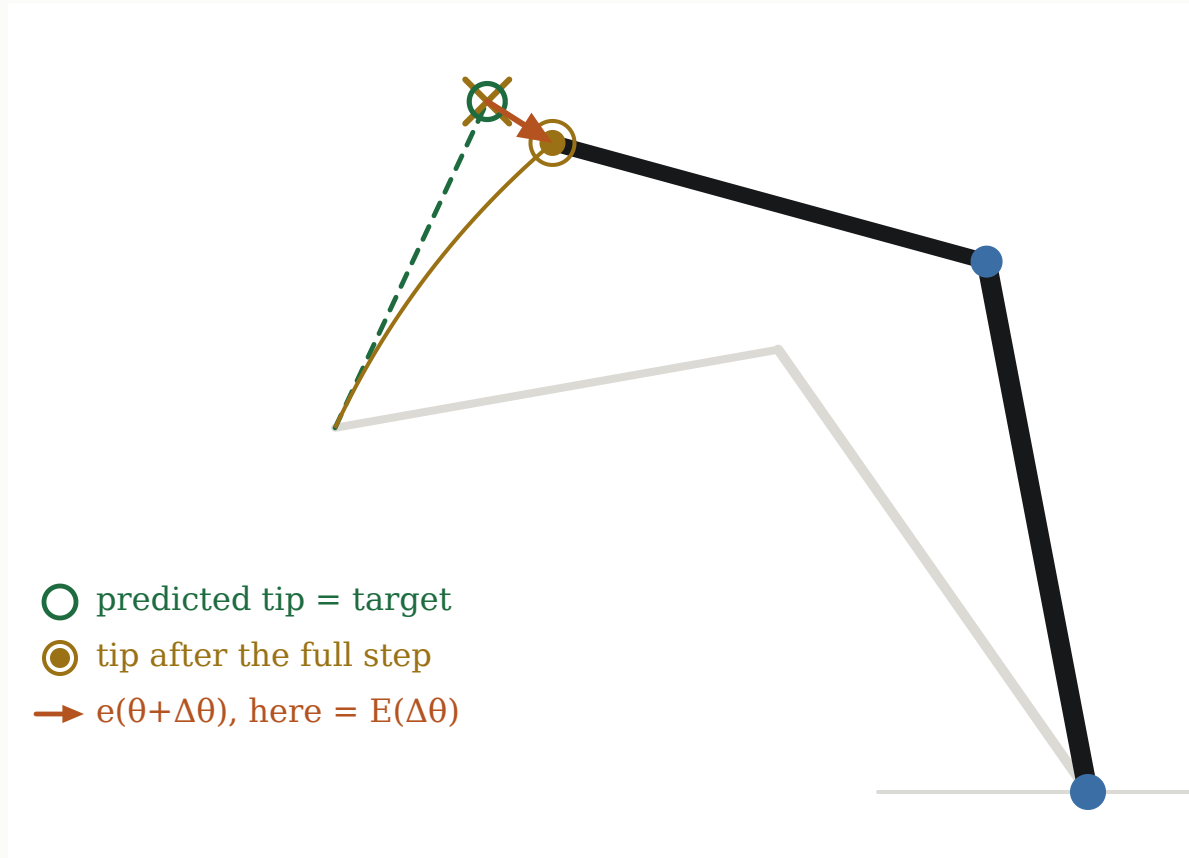
$$\min_{\Delta\theta} \|J\Delta\theta + e\|^2$$

$J\Delta\theta + e$ the model's error after the step

J full column rank here

Robotics Application

Predicted Motion and Actual Motion



$$e(\theta) + J\Delta\theta = 0 \text{ but } e(\theta + \Delta\theta) \neq 0$$

in this example

$e(\theta) + J\Delta\theta$ the model's error after the step; 0 here

$e(\theta + \Delta\theta)$ the true error after moving

$E(\Delta\theta)$ their difference: model discrepancy,
 $O(\|\Delta\theta\|^2)$

Zero linearised error need not mean zero true error.

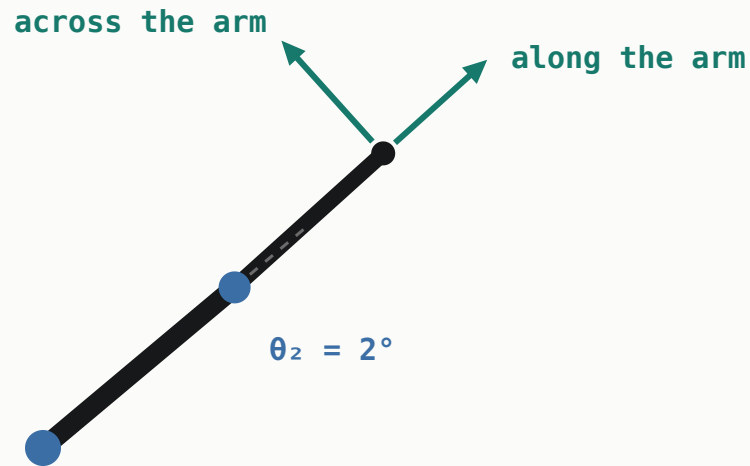
Linearise, step, measure, relinearise, repeat — each step small enough that the linear model stays accurate.

Robotics Application

Ill-Conditioned or Singular?

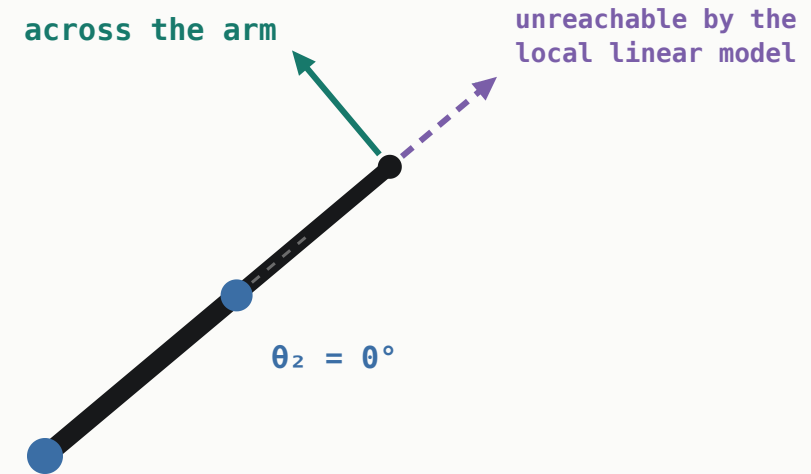
$$\det J = l_1 l_2 \sin \theta_2$$

l_1, l_2 link lengths; θ_2 elbow angle



$\det J \neq 0$: full rank, R invertible

QR runs; step size depends on e 's direction



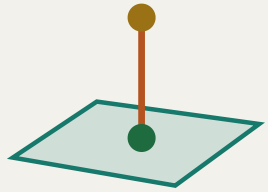
$\det J = 0$: rank 1, $r_{22} = 0$

Ordinary solve fails, yet minimisers exist; no joint rate gives along-arm tip velocity

Synthesis and Exit Ticket

One Problem, Four Views

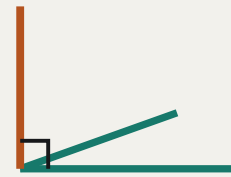
$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$



GEOMETRY any A

$$\hat{b} = \text{proj}_{\text{Col}(A)} b$$

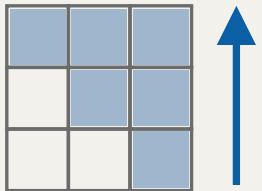
the closest reachable output



ALGEBRA any A

$$A^T A \hat{x} = A^T b$$

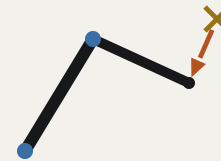
the equations every minimiser satisfies



COMPUTATION full column rank

$$A = QR$$

the minimiser satisfies $R\hat{x} = Q^T b$



ROBOTICS

full column rank · local model

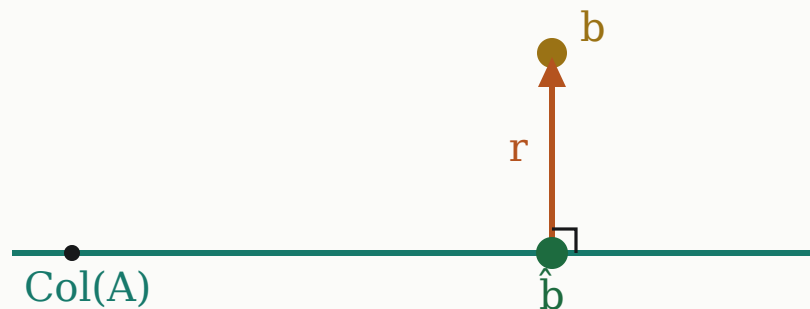
$$R\Delta\theta = -Q^T e$$

one step, then relinearise

Synthesis and Exit Ticket

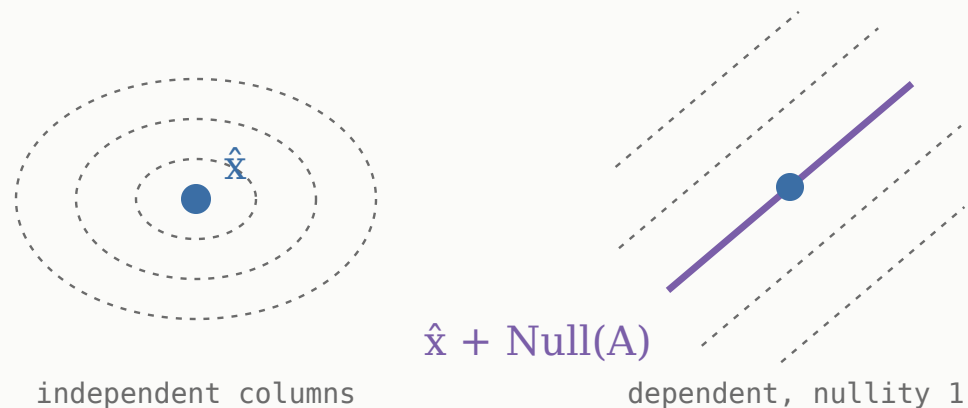
One Best Output, Sometimes Many Parameter Vectors

measurement space $\mathbb{R}^m$



Every A and b : one closest $\hat{b}$, one $r \perp \text{Col}(A)$.

parameter space $\mathbb{R}^n$



$\{x : Ax = \hat{b}\} = \hat{x} + \mathcal{N}(A)$: one point exactly when the columns are independent.

Projection: no assumption. $R\hat{x} = Q^T b$ and $(A^T A)^{-1}$: full column rank.

Synthesis and Exit Ticket

Find the Incorrect Claim

Which claim is false? Sketch $\text{Col}(A)$ and b so it fails.

(A) $r = b - A\hat{x}$ is $\perp$ each column of A .

(B) If $b \in \text{Col}(A)$, the minimum residual is 0.

(C) A nonzero minimum residual forces a non-unique $\hat{x}$.

(D) Full column rank makes $\hat{x}$ unique.

		full column rank?	
		yes	no
$b \in \text{Col}(A)?$	yes		
	no		

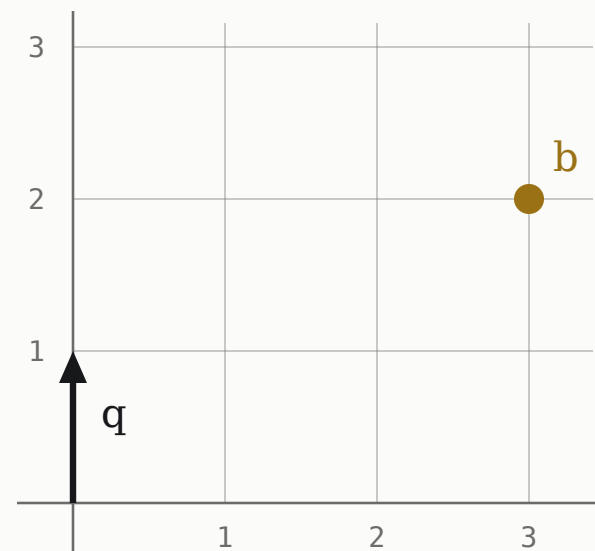
Synthesis and Exit Ticket

Exit Ticket

1. $q = (0, 1)$, $b = (3, 2)$. Projection p ? Residual r ?
2. Why does $A^T r = 0$ not force $r = 0$? **One sentence of reasoning.**
3. Thin QR, full rank: which part of $\|QRx - b\|^2$ can no x change?

 → **Hand in before the answers.**

Nearly dependent or rank-deficient columns: next lecture.



Courses and books

1. Axler, S. [*Linear Algebra Done Right*](#), 4e, 6A–6C. Springer, 2024.
2. Strang, G. [*18.06 Linear Algebra*](#), L14–L17. MIT OCW, 2010.
3. Trefethen & Bau. *Numerical Linear Algebra*, Lec. 6–8, 10–11, 18–19. SIAM, 1997.
4. Boyd & Vandenberghe. [*Introduction to Applied Linear Algebra*](#), 5.4, 10.4, 12–13, 18. Cambridge, 2018.
5. Lynch & Park. [*Modern Robotics*](#), 5.3, 6.2. Cambridge, 2017.

Course standards: [*Notation and Conventions*](#), symbols, shapes and the two sign conventions; [*Textbooks and Reading Guide*](#), what each source is for and where its notation differs.